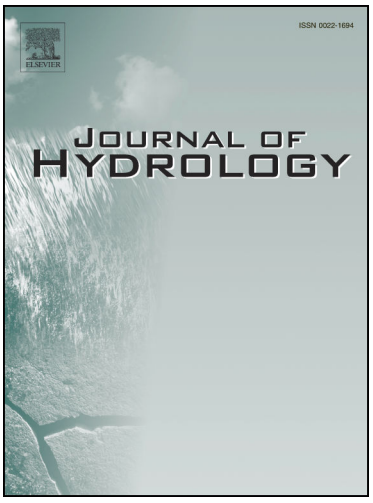
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Soil-crop hydrology modelling approaches for conjunctively estimating recharge, crop demand and pumping in irrigated agriculture: a review

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Soil-crop hydrology modelling approaches for conjunctively estimating recharge, crop demand and pumping in irrigated agriculture: A review

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Highlights

- 16 models evaluated for crop, soil hydrology and irrigation simulation.
- 13 models simulate water stress; only 7 also represent salinity stress.
- Only 4 models trigger irrigation using crop-stress factors.
- Modelled pumping was compared with metered data in only 2 studies.
- Runoff was reconciled with field data in only 1 of 13 case studies.

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Abstract

Soil-crop hydrology models are frequently used to estimate regional-scale groundwater recharge, crop water use, runoff and freshwater extraction rates to support irrigation demands. Because interactions between soils, crops and irrigation practices influence recharge, crop water requirements and pumping, it is important to apply models that can simulate these processes simultaneously. This review examines 16 soil-crop hydrology models and 13 case study applications in which recharge, irrigation demand and extraction rates (i.e., from surface water and/or groundwater) were estimated to support regional water accounting and water-resource investigations. Model functions for runoff, soil water movement, root water uptake, crop growth and irrigation were compared to identify differences between models and their limitations. This review showed that 13 of the 16 models incorporate crop growth simulation linked to soil water dynamics, while 7 also account for salinity stress. Twelve models allow irrigation to be triggered based on soil water content, with 4 also allowing irrigation to respond to crop stress factors. The reliability of soil-crop hydrology models for estimating pumping rates remains uncertain because pumping was validated against metered data in only 2 of the 13 case studies reviewed. In those studies, metered extraction data provided important constraints on model parameterisation. Although runoff was simulated, it was reconciled with field observations in only 1 of the 13 case studies. Soil-crop hydrology models remain important tools for estimating recharge and irrigation demand, however, the representation of conjunctive use of groundwater and surface water, and the resulting pumping estimates, are commonly imposed externally and remain weakly validated against field observations.

Keywords: Water resources, Soil-crop hydrology models, Unsaturated zone, Groundwater management, Salinity, Crop growth

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1 Introduction

Global demands for agricultural production are increasing, leading to higher rates of irrigation water use from shrinking freshwater resources (De Graaf et al., 2019; Qureshi et al., 2013b). In meeting these demands, the management of surface and subsurface freshwater in agricultural areas requires reliable estimates of infiltration, groundwater recharge, pumping rates, surface runoff, crop water demands and irrigation practices. However, there are often gaps in the monitoring data needed to achieve accurate accounting of surface water and groundwater balances. For example, the metering of water extraction is often lacking, especially for groundwater, and yet, pumping rates are important inputs in managing human-imposed stresses on aquifers (Cheng et al., 2009). More generally, the spatial and temporal variability of both pumping and recharge need to be characterised for the creation of groundwater models and to develop regional groundwater management strategies (Knowling et al., 2015).

Rainfall and irrigation recharge to groundwater are key variables in water resource investigations of irrigated areas, and are commonly determined using soil-crop hydrology models. In this review, soil-crop hydrology models refer to vertically resolved (typically 1D) models that simulate water balance processes within the soil profile coupled with crop water uptake and growth processes (e.g., Sun et al., 2015; Ventrella et al., 2019). Although simulations typically occur in 1D, spatially distributed application using GIS-derived inputs enables regional-scale estimation of crop water uptake, recharge and irrigation demand, and associated pumping rates (e.g., Morgan et al., 2015; Stöckle et al., 2014; Winchell et al., 2010).

Soil-crop hydrology models allow rainfall to be partitioned into evapotranspiration, runoff, surface infiltration and recharge based on inputs of meteorological data, vegetation parameters, soil hydraulic properties, and irrigation and land-use practices (Keese et al., 2005). In cases where the water table influences the soil water content, groundwater conditions may also need to be included. Soil-crop hydrology models have additionally been used to estimate crop water demand and shortfalls in the available soil water content to meet crop needs (Ahmad et al., 2002; Deines et al., 2021; Hyndman et al., 2017). As such, these models can be used to estimate irrigation requirements. In addition to the biophysical elements described above, the simulation of irrigation applications often requires representation of farming practices, including triggered irrigation, where water is added to the soil in response to the depletion of soil water content, increased soil salinity, and/or identified crop needs (e.g., Brisson et al., 2009; Neitsch et al., 2011). Estimates of recharge and the other fluxes described above that are simulated by soil-crop hydrology models require independent substantiation (e.g., Okrah et al., 2023; Ordens et al., 2014; von Freyberg et al., 2015). This is consistent with the recommendations of Scanlon et al. (2002) and Becke et al. (2024), who highlight the importance of using multiple techniques to estimate recharge given the high uncertainty in commonly adopted methods.

The parameters adopted in soil-crop hydrology models that represent the soil type, meteorological stresses, land use, agricultural practices, and lower-boundary drainage conditions vary between 1D model simulations so that the spatial variability of the study area

can be captured. Individual simulations representing discrete regions within an area of interest are then combined, and the resulting fluxes are aggregated to represent conditions at regional scales (e.g., Reading et al., 2019). This produces soil-crop water budgets that are integrated over management areas or to a spatial resolution needed for groundwater and/or surface water modelling. The conversion of crop demand and rates of irrigation from 1D models into water extraction rates requires upscaling to cropped areas, accounting for the locations of pumping wells, surface water diversion or intake points, and the land areas to which irrigation water is supplied from each well, river intake or channel offtake. The pumping rates obtained from this approach are important for characterising the spatial distribution of groundwater and surface water stresses that are routinely the targets of water management investigations and important inputs to groundwater and surface water models (Yang et al., 2010).

The current study builds on previous reviews of soil-crop hydrology models by focusing on 1D modelling approaches that combine the estimation of crop water uptake, irrigation demand, pumping rates and recharge rates at the catchment scale. Previous reviews include the work of Connolly (1998), who summarised the functions in soil-crop models in terms of soil structural properties and their influence on hydrological processes (water balance components) in soil-crop systems. Connolly (1998) concluded that developments were needed to extend point-scale models to better capture spatial variability at the catchment scale, amongst other recommendations related to the representation of soil hydraulic properties. Kanda et al. (2018) reviewed methods for coupling hydrological and crop models, demonstrating the potential benefits arising from incorporating crop models in hydrological simulation for water resources conservation, sustainability and agricultural performance improvement. Tenreiro et al. (2020) also reviewed seven crop models and five hydrology models to assess their capabilities for simulating spatial variability of soil water movement at the local (farm) scale. Tenreiro et al. (2020) concluded that there were opportunities to improve both crop-focused models and hydrology models to incorporate spatial variations at the local scale, which appeared to have a significant influence on soil water behaviour. Vereecken et al. (2016) analysed the role of soil modelling in quantifying ecosystem services. They found that even advanced soil models, which incorporate root water uptake and crop growth, still face considerable challenges in accurately capturing the spatial and temporal dynamics of soil water movements and nutrient availability, and their interactions with plant roots.

Wang and Smith (2004) reviewed modelling approaches for the growth and water uptake of plants, showing that more detailed consideration of root distribution and responses to soil conditions are needed. Dhakar et al. (2018) summarised how various crop models incorporate crop water stress and simulate its impact on different growth parameters and yield. They concluded that crop models often oversimplify crop water stress and its effects, leading to uncertainty in simulating growth and yield, especially under water-limited conditions. Uniyal and Dietrich (2021) reviewed soil-crop hydrology modelling tools that can be applied to the estimation of irrigation demand at the regional scale. They analysed the irrigation scheduling functions in 14 soil-crop hydrology models and provided a basic comparison of soil water movement and plant growth approaches. Uniyal and Dietrich (2021) concluded that soil-crop hydrology models hold significant potential for optimising irrigation management. They also identified key limitations, including inadequate support for important crops such as rice, limited representation of agroforestry systems, and the insufficient integration of water-saving strategies such as deficit irrigation, water reuse and advanced technological innovations.

Examples of the latter include daily real-time irrigation forecasting at the catchment scale using remote sensing technologies and the application of meteorological forecasting to provide reliable soil moisture predictions.

None of the prior reviews of soil-crop hydrology models have evaluated the modelling approaches for conjunctively estimating recharge, crop water demand and water extraction, including the partitioning of irrigation demand between surface water and groundwater sources. These applications of soil-crop hydrology models are vital for capturing the complex interactions between irrigation practices, soil and aquifer dynamics and other catchment water balance components, such as runoff and crop water use, which are fundamental considerations for sustainable water resources management.

The current review aims to evaluate the key model functions and conceptual elements of existing soil-crop hydrology modelling tools applied to the coupled estimation of recharge and irrigation water supply rates (i.e., leading to estimates of surface water and groundwater extraction). The review includes 8 case studies to assess how these tools are applied in various real-world contexts. The outcomes of this review are expected to inform hydrological investigations of agricultural areas, where the coupled simulation of soil hydrology, crop water use, irrigation practices and recharge is necessary to inform water management decision making. In this way, the current review builds on prior articles by Dhakar et al. (2018), Tenreiro et al. (2020), Uniyal and Dietrich (2021), and Wang and Smith (2004), which explored soil-crop hydrology models from related but different perspectives.

2 Conceptual model of rainfall partitioning, crop demand, soil water movement and irrigation

The soil-crop hydrology models that are used to estimate recharge and irrigation application rates need to account for numerous surface and subsurface hydrological processes. These encompass several interconnected physical components, such as the vegetation canopy, land surface, root zone, unsaturated zone and, often, the water table. The models parameterise these components and apply water balance and flux equations to partition rainfall and irrigation inputs into evaporation (from soil and interception evaporation), transpiration, infiltration, surface runoff, root water uptake, deep drainage and groundwater recharge, as depicted in Figure 1.

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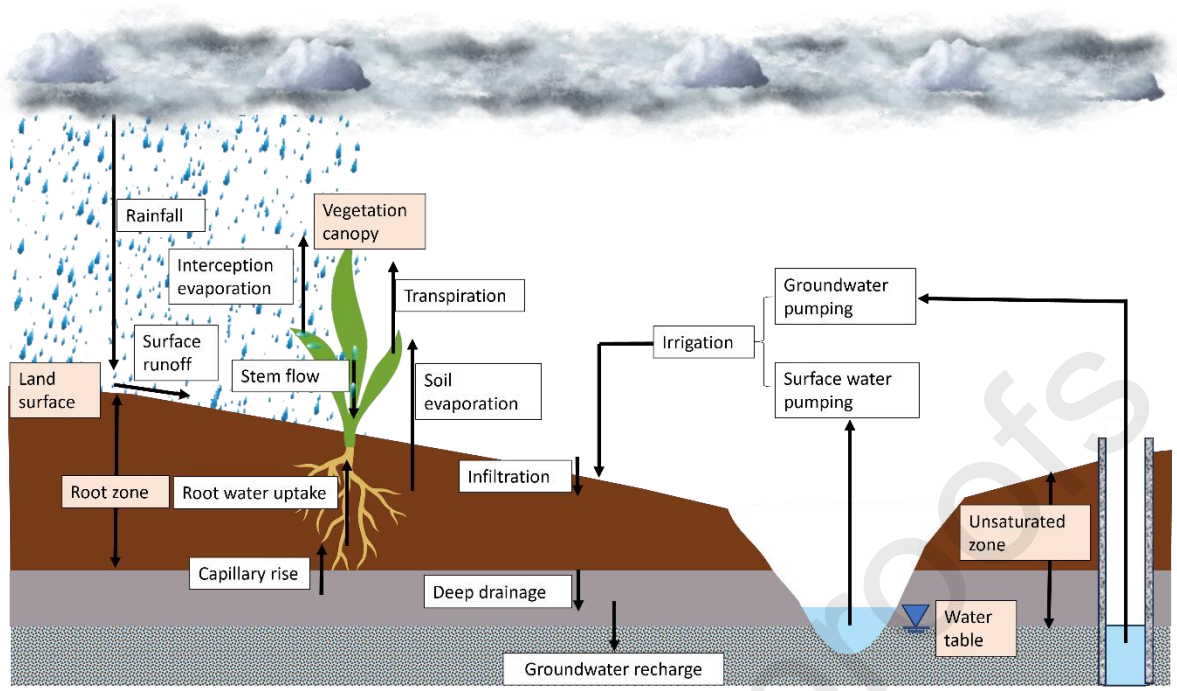


Figure 1. Conceptual model of the partitioning of rainfall and irrigation into interception, stem flow, evaporation, transpiration, infiltration, surface runoff, root water uptake, deep drainage and recharge. Orange-shaded boxes are components, while white-shaded boxes are fluxes. Throughfall, representing rainfall that bypasses the canopy directly to reach the soil surface, is not shown separately from stem flow. Shallow water table conditions are depicted, whereas in some settings, the root zone does not extend to the top of the capillary fringe. The channel shown schematically represents a surface-water source for irrigation.

Water inputs to soil-crop hydrology models include precipitation (most commonly rainfall) and irrigation. The former is obtained from meteorological datasets and is treated as spatially varying for regional-scale applications (e.g., Reading et al., 2019). Soil-crop hydrology models apply various methods to determine the timing and application rates of irrigation. Irrigation timing is regulated through user-defined schedules or may be triggered based on the soil water content, crop health indicators or rainfall deficits (e.g., Hutson, 2003; Neitsch et al., 2011; van Dam et al., 1997). The irrigation amount is typically governed by predefined thresholds, with water applied up to field capacity or according to user-specified values (e.g., Hutson, 2003; Neitsch et al., 2011; van Dam et al., 1997).

Rainfall and irrigation applied to the vegetation canopy are generally partitioned into vegetation interception, stem flow and throughfall, where throughfall refers to rainfall that bypasses the canopy and reaches the soil surface directly (Dunkerley and Booth, 1999). Vegetation interception is the retention of water on the plant surface and is generally assumed to return to the atmosphere via evaporation (Dunkerley, 2000). The rates of interception depend on canopy storage, interception (or canopy) evaporation, stem flow and throughfall, and may be calculated in soil-crop hydrology models using several alternative representations. For example, evaporation losses arising from interception may be approximated using a simple loss term, or a canopy storage model may be used to assess interception based on some measure of

the canopy mass, such as the leaf area index (LAI) (Neitsch et al., 2011; van Dam et al., 1997), which is the one-sided leaf area per unit of ground area (Zheng and Moskal, 2009).

Water reaching the land surface is further partitioned into runoff and infiltration, along with the storage of ponded water at the land surface in depressions or as microtopographic rills storage, allowing for delayed infiltration and acting to retard surface runoff to streams. When the movement of runoff between modelled regions is known, runoff to downslope areas can be treated as "run-on", which adds to the water inputs to downslope areas, thereby enhancing infiltration, soil water content and runoff in lower topographical regions (Wang and Chen, 2021). Run-on is often ignored in the application of 1D hydrology models (Nahar et al., 2004), although it is considered in some models, such as SWAP (van Dam et al., 1997).

Infiltrated water enters the root zone, where it may be taken up by plant roots or lost to direct evaporation. Root water uptake in soil-crop hydrology models is calculated as a function of the crop type and stage of development, and the soil water content, soil hydraulic properties and atmospheric factors (e.g., Jones et al., 2003; Stöckle and Nelson, 2003). The functions used for root water uptake reflect the main ecohydrological factors that drive transpiration. Root water uptake is typically assumed to equal transpiration because plant water storage is not represented in the models evaluated in the current study. Transpiration rates may be approximated as a simple proportion of the potential evapotranspiration (e.g., Hutson, 2003), where the latter is most commonly calculated using the Penman-Monteith equation (Lang et al., 2017; Monteith, 1965). Crop factors and other multipliers are applied to the potential evapotranspiration to account for local conditions (e.g., Stöckle et al., 2003). The soil water losses arising from transpiration and direct evaporation are often partitioned so that plant water use can be quantified and, in some models, plant health can be assessed (Wei et al., 2015). This is accomplished using various methods. For example, HYDRUS-1D uses Beer's law to partition the solar radiation component of the energy budget into potential plant transpiration and potential soil evaporation using the soil cover fraction, which is derived from the LAI (Šimůnek et al., 2013).

The excess water that drains below the root zone is considered deep drainage. If the deep drainage is presumed to be groundwater recharge, the time lag between the onset of deep drainage and the arrival of this water at the water table is neglected (Timms et al., 2012). Otherwise, a lower water table boundary condition is needed that reflects local conditions to properly investigate the timing of recharge (e.g., Ordens et al., 2014). This requires a higher computational burden (i.e., in numerical models) for regions with deep water tables, which are more common in arid and semi-arid regions, and mountainous areas (Nazarieh et al., 2018). Capillary rise from the water table may add to the inputs to the unsaturated zone of soil-crop hydrology models and can be an important influx under shallow water table conditions (Ordens et al., 2014). The simulation of capillary rise requires a lower boundary condition at the base of the unsaturated zone that mimics the water table.

3 Model functions relevant to the joint estimation of groundwater recharge and pumping

Following the conceptual model of rainfall partitioning, soil-crop hydrology models provide a numerical framework for quantifying the individual components of the water balance and their interactions with irrigation management. The reliability of groundwater recharge and irrigation water extraction estimates depends on the representation of key hydrological and crop processes, including runoff generation, infiltration and soil water redistribution, root water uptake and evapotranspiration, crop growth, irrigation scheduling, and boundary conditions (e.g., Gandolfi et al., 2006; Wang and Smith, 2004; Zheng et al., 2017). Therefore, this section evaluates the major model functions that influence the simulation of groundwater recharge, irrigation demand, and associated water extraction.

3.1 Surface runoff estimation

Surface runoff plays a crucial role in rainfall partitioning and, consequently, in the estimation of recharge and soil moisture content (Weatherl et al., 2021). Surface runoff induced by the application of water to the land surface is typically subdivided into: (a) infiltration excess, which occurs when inputs to the land surface exceed the infiltrability of the surface soil, and (b) saturation excess, which occurs when the soil profile becomes saturated, sometimes associated with the water table reaching the land surface (Dunne and Black, 1970). Here, the infiltrability is the infiltration flux, or rate of water entry into the soil, when water at atmospheric pressure is freely available at the land surface (Hillel, 1980). The determination of infiltration excess runoff requires estimation of the soil water content and the accompanying unsaturated hydraulic conductivity (K_u [LT^{-1}]), whereas estimation of saturation excess runoff may require the groundwater conditions to be characterised if high water table conditions are the cause (Stewart et al., 2019). Additionally, infiltration excess runoff and saturation excess runoff interact with each other, which further adds to the complexity of runoff estimation. Most of the models evaluated in the current review do not simulate groundwater flow processes and water table changes explicitly, and therefore saturation-excess runoff arising from the water table reaching the land surface is not generally determinable through these models. Rather, surface runoff is estimated using empirical functions that do not differentiate between infiltration excess and saturation excess runoff (as discussed below).

Most soil-crop hydrology models apply the Curve Number (CN) method (NRCS, 2004) for the estimation of surface runoff (Table 1). The CN method estimates direct runoff from rainfall by considering the soil type, land use and cover type (e.g., crop, forest, agricultural lands, etc.), land treatment (i.e., conservation practices), topography, and the antecedent soil water content (NRCS, 2021). The values of the input parameters for the CN method have been derived in previous studies primarily from observations of rainfall and streamflow (e.g., Moglen et al., 2022). As such, the CN method does not necessarily distinguish between different runoff-generation mechanisms, but instead provides an empirical estimate of catchment-scale runoff based on observations of rainfall and stream discharge (NRCS, 2004). This presents challenges

where soil-crop hydrology models adopting the CN method are used for process-based simulation of catchment water balances because different overland flow processes, such as run-on, infiltration excess runoff and saturation excess runoff, are not distinguished in the CN method and thereby are not explicitly represented within the model water balance (Garen and Moore, 2005).

Table 1. Runoff simulation approaches adopted in soil-crop hydrology models. Models 1-14 are primarily 1D soil-crop hydrology models, while models 15 and 16 (marked with "*") are semi-distributed catchment models. Where there was no evidence in the user manual or key papers that a given approach is available in the model, "-" is used.

No.	Model	Runoff simulation approach**	
		Curve number method	Other approaches
1	APSIM	Yes[1]	-
2	LEACHM	Yes[2]	Infiltration excess runoff (where potential infiltration exceeds the soil's infiltration capacity)[2]
3	SPLASH	-	User-defined runoff filter[3]
4	SWAP	-	Linear relationship between ponding depth and surface runoff, via a resistance term (typically 0.1 d)[4]
5	CropSyst	Yes[5]	-
6	HYDRUS-1D	Yes[6]	Infiltration excess runoff[6]
7	SALUS	Yes[7]	-
8	RZWQM2	-	Green-Ampt[8] infiltration method and infiltration excess runoff[9]
9	DSSAT	Yes[10]	-

10	STICS	-	Simple proportion of rainfall above a given threshold[11]
11	AquaCrop	Yes[12]	-
12	Daisy	-	Simple linear model based on a ponding threshold[13]
13	EPIC	Yes[14]	-
14	WAVES	-	Infiltration excess runoff and saturation excess runoff (upper boundary switched to a specified-head boundary under saturated conditions)[15]
15	MIKE SHE*	-	Threshold water depth with numerical flow calculations[16]
16	SWAT*	Yes[17]	Green-Ampt[8] infiltration method[17]

[**]Sources: (1) Vogeler et al., 2023; (2) Hutson, 2003; (3) Arunakumaren, 1997; (4) van Dam et al., 1997; (5) Stöckle and Nelson, 2003; (6) Hilten et al., 2008; (7) Basso et al., 2006; (8) Green and Ampt, 1911; (9) Ma et al., 2012; (10) Jones et al., 2003; (11) Brisson et al., 2009; (12) Raes et al., 2009; (13) Hansen, 2002; (14) Sharpley and Williams, 1990; (15) Zhang and Dawes, 1998; (16) DHI, 2024; (17) Neitsch et al., 2011.

3.2 Soil water movement

The two methods used to simulate water movement and storage in the unsaturated zone within soil-crop hydrology models are: (a) tipping-bucket models or cascading/capacity models (Ritchie, 1985), and (b) Richards' equation (Richards, 1931). Both methods allow for the determination of the soil water content (θ [-]) and estimates of vertical soil water fluxes (q [LT^{-1}]) as a function of depth, while Richards' equation also requires consideration of the soil water potential (Ψ [L]) and provides a physically based estimate of soil water fluxes (Emerman, 1995), as discussed below. In both approaches, the water content is often given in terms of the degree of saturation, S [-], defined as:

$$S = \frac{\theta - \theta_r}{\theta_s - \theta_r} \quad (1)$$

where θ_r [-] is the residual soil water content and θ_s [-] is the saturated soil water content.

316 Rigby and Porporato (2006) and Romano et al. (2011) present water balance equations for
317 tipping-bucket models, which can be written in simplified form as:

318 $$\frac{\partial W}{\partial t} = I - E - T - L \quad (2)$$

319 where W [L] is the water stored in each bucket expressed as equivalent water depth (e.g., $W = \theta Z$, where Z [L] is the bucket thickness), I [LT^{-1}] refers to infiltration, E [LT^{-1}] indicates
321 evaporation loss, T [LT^{-1}] is transpiration loss, L [LT^{-1}] is the leakage rate through the base of
322 each bucket, and t [T] is time. In multi-bucket simulators, L is the net leakage rate, representing
323 the difference between incoming leakage from any overlying bucket and the loss to the bucket
324 below. The I, E and T terms apply principally to the top bucket, depending on the distribution
325 of vegetation roots and soil bucket depths. The L term may be formulated in various ways, but
326 generally takes the form:

327 $$L = K_d(\theta - \theta_u) \quad \theta > \theta_u$$
$$L = 0 \quad \theta \leq \theta_u \quad (3)$$

328 where K_d [LT^{-1}] is a vertical conductance parameter (i.e., a drainage rate constant) that is
329 commonly related to the vertical hydraulic conductivity K_v [LT^{-1}], and θ_u [-] is the water content
330 at which downward drainage is presumed to reduce to an insignificant rate, often presumed to
331 be the field capacity (FC).

332

333 The tipping-bucket model of SALUS (Basso et al., 2012) does not require explicit estimates of
334 hydraulic conductivity, which can be an advantage given challenges in its field characterisation
335 (Fodor et al., 2011). However, in SWAT, the saturated hydraulic conductivity (K_s [LT^{-1}]) may
336 be used to estimate the travel time between buckets as $Z(\theta_s - \theta_u)/K_s$, where Z [L] is the bucket
337 thickness, and θ_u [-] is taken to be the FC. The travel time is used to limit the water movement
338 between soil buckets within the model time step (Neitsch et al., 2011).

339

340 In tipping-bucket models (e.g., APSIM-SoilWat (Keating et al., 2003), DSSAT (Jones et al.,
341 2003), CropSyst (Stöckle and Nelson, 2003)), soil water characteristics for each layer are
342 defined in terms of the permanent wilting point (PWP), the drained upper limit (DUL) and θ_s.
343 The PWP is the lowest water content from which plants can extract water, while the DUL is
344 the upper limit of plant-available water after drainage has ceased, approximately equal to the
345 FC (Veihmeyer and Hendrickson, 1949). Thus, DUL is approximately equal to the water
346 content θ_u at which downward drainage becomes negligible.

347

348 A limitation of many tipping-bucket models is the inability to simulate upward water
349 movement, including capillary rise, which contributes to surface evaporation and sustains root-
350 zone water during dry periods depending on the soil type and water table depth (Turek et al.,
351 2022). Where this process is excluded, it may lead to an underestimation of the plant-available
352 water under shallow water table conditions (Jarvis et al., 2022).

353

Richards' equation (Richards, 1931) is a partial differential equation that describes the movement of water through variably saturated soil. It is solved using numerical methods, such as finite-difference and finite-element approaches (Feddes et al., 1988), providing θ and Ψ at each node. Soil water fluxes are determined through Darcy's law using unsaturated zone hydraulic properties (Pachepsky et al., 2003). Richards' equation for vertical flow is given as (Feddes, 1978; Richards, 1931):

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z}\left[K_u\left(\frac{\partial \Psi}{\partial z} - 1\right)\right] - U_{z,t} \quad (4)$$

where z [L] is the vertical coordinate taken as positive upward, and U [T^{-1}] is a sink term representing water influxes and losses that may vary with depth and in time. Here, $\theta = \theta(\Psi)$ defines the water retention relationship between matric potential and water content, while $K_u = K_u(\theta)$ or $K_u = K_u(\Psi)$ defines the hydraulic conductivity function, which is coupled to the moisture retention relationship through water content or matric potential (Hillel, 2004). Pedotransfer functions are commonly adopted in parameterising the water retention properties and hydraulic conductivity, whereby soil physical properties (e.g., particle size distribution, bulk density and organic carbon content) are translated into constants within $\theta(\Psi)$ and $K_u(\theta)$ functions, such as those defined by Brooks and Corey (1964), Campbell (1974), van Genuchten (1980), and others.

Comparisons of tipping-bucket models and Richards' equation have been undertaken in several studies (e.g., Emerman, 1995; Kholis et al., 2025). Nolan et al. (2005) concluded that better results might be gained from tipping-bucket models when there are insufficient data for the parameterisation of Richards' equation. However, Emerman (1995) highlighted the limitation of most tipping-bucket models in that the timestep is commonly fixed, usually daily, and yet sub-daily variability in soil moisture, evapotranspiration, etc. may be important in some contexts. Thus, tipping-bucket models usually cannot take advantage of higher temporal resolution data or simulate processes requiring shorter timesteps.

The occurrence of soil cracks and other preferential flow pathways (e.g., solution channels, macropores, root-induced channels) may be important factors in controlling soil water movements (Du et al., 2025; Lu et al., 2020). These allow water to move rapidly downward, referred to here as macropore or bypass flow, rather than infiltrating uniformly into the soil matrix (Novák et al., 2000). Additional equations are needed to simulate macropore flow in Richards' equation models, which are intended to represent capillary-driven water movement through small pores rather than gravity-driven macropore flow (Beven and Germann, 2013). Example functions for macropore flow are given by Beven and Germann (2013), Emerman (1995), Šimůnek and van Genuchten (2008), and van Dam et al. (2008). Tipping-bucket models represent macropore flow using simplified or empirical formulations. For example, Poon et al. (2021) modified the tipping-bucket model approach of the SWAT-MAC model to account for macropore flow by dividing each soil layer into a matrix domain and a macropore domain, with the rate of macropore flow determined by the amount of daily infiltration that exceeds the infiltration capacity of the matrix domain, scaled by a dimensionless macropore connectivity factor. The latter depended on the soil moisture content, thereby reflecting changes in soil cracks with moisture content. The model functions applied to simulate soil water movement in the reviewed models are compared in Table 2.

Table 2. Model functions applied in simulating soil water movement. Models 1-14 are primarily 1D soil-crop hydrology models, while models 15 and 16 (marked with "*") are semi-distributed catchment models. Where no evidence was found in the user manuals or key papers that a given approach was adopted, "-" is used.

No.	Model	Tipping-bucket model**	Richards' equation model**	Hydraulic functions**		Built-in pedotransfer function**	Macropore flow**
				Water retention function	Hydraulic conductivity function		
1	APSIM-SWIM	-	Yes[1]	Brooks and Corey (1964) function and van Genuchten (1980) function[1]	Mualem (1976) function[1]	Yes[1]	-
	APSIM-SoilWat	Yes[2]	-	-	-	-	-
2	LEACHM	Yes[3]	Yes[3]	Campbell (1974) function, Hutson and Cass (1987) function, van Genuchten (1980) function[3]	Campbell function (1974), Hutson and Cass (1987) function, van Genuchten (1980) function[3]	Yes[3]	Yes[4]
3	SPLASH	Yes[5]	-	-	-	-	Yes[5]
4	SWAP	-	Yes[6]	van Genuchten (1980) function[6]	van Genuchten (1980)-Mualem (1976) function[6]	Yes[6]	Yes[6]

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5	CropSyst	Yes[7]	Yes[7]	Campbell (1974) function[7]	Campbell (1974) function[7]	Yes[8]	Yes[7]
6	HYDRUS-1D	-	Yes[9]	Brooks and Corey (1964) function, van Genuchten (1980) function, Vogel and Císlerová (1988) function, Kosugi (1996) function, and Durner (1994) function[10]	Brooks and Corey (1964) function, van Genuchten (1980)-Mualem function (1976), Vogel and Císlerová (1988) function and Durner (1994) function combined with Mualem (1976) function[10]	Yes[11]	Yes[12]
7	SALUS	Yes[13]	-	-	-	-	-
8	RZWQM2	-	Yes[14]	Slightly modified Brooks and Corey (1964) function[14]	-	-	Yes[14]
9	DSSAT	Yes[15]	-	-	-	-	-
10	STICS	Yes[16]	-	-	-	-	Yes[16]
11	AquaCrop	Yes[17]	-	-	-	Yes[18]	Yes[17]
12	Daisy	-	Yes[19]	van Genuchten (1980) function, Brooks and Corey (1964) function, and Campbell (1974) function[19]	Burdine (1953) or Mualem (1976) functions combined with selected retention function[19]	-	Yes[19]

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13	EPIC	Yes[20]	-	-	-	-	Yes (crack-flow)[20]
14	WAVES	-	Yes[21]	-	-	-	-
15	MIKE SHE*	-	Yes[22]	van Genuchten function (1980) and Campbell function (1974)[22]	Averjanov (1950) function, van Genuchten (1980) function and Campbell (1974) function and Burdine function (1953)[22]	-	Yes[22]
16	SWAT*	Yes[23]	-	-	-	-	Yes[23]

⁂Sources: (1) Verburg et al., 1996; (2) Holzworth et al., 2014; (3) Hutson, 2003; (4) Hutson and Wagenet, 1995; (5) Arunakumaren, 1997; (6) van Dam et al., 1997; (7) Stöckle and Nelson, 2003; (8) Stöckle et al., 2003; (9) Lazarovitch et al., 2023; (10) Šimůnek et al., 2013; (11) Hilten et al., 2008; (12) Šimůnek et al., 2023; (13) Basso and Ritchie, 2015; (14) Ma et al., 2012; (15) Jones et al., 2003; (16) Brisson et al., 2009; (17) Pourgholam-Amiji et al., 2021; (18) Raes et al., 2009; (19) Hansen, 2002; (20) Williams and Renard, 1985; (21) Dawes, 1998; (22) DHI, 2024; (23) Neitsch et al., 2011.

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Table 2 shows that soil water movement is simulated using either tipping-bucket models or Richards' equation, with variation among models in the representation of hydraulic functions, pedotransfer functions and representation of macropore flow. Richards' equation models generally provide more detailed representation of hydraulic functions, pedotransfer options and macropore flow than tipping-bucket models, although macropore flow is also represented in some tipping-bucket approaches. APSIM, LEACHM and CropSyst are the only models, among those considered, that provide both tipping-bucket and Richards' equation approaches for simulating soil water movement.

3.3 Root water uptake

Root water uptake is driven by transpiration and involves complex flow processes occurring within the soil-plant-atmosphere continuum (Feddes et al., 2001; Hillel, 2004). Soil hydraulic properties and the distribution of roots determine the plant's access to soil water (Feddes et al., 2001), while atmospheric evaporative demand together with plant size and canopy structure drive transpiration (Kim et al., 2014). Water moves to the roots via the root-soil interface and into the xylem, where it is transported upward through stems to the leaves. In the leaves, water evaporates from moist cell surfaces into the intercellular spaces, diffuses through the stomata, and then passes into the atmosphere. During transpiration, water pressure in the leaves decreases below atmospheric levels, creating a tension that drives the upward flow of water from roots to leaves through the xylem vessels (Kim et al., 2014). A plant can sustain normal physiological function provided that, over sustained periods, transpiration demand does not exceed the capacity of the root system to supply water; otherwise, the leaf water potential drops, leading to loss of turgor (reduction in hydrostatic pressure within plant cells), stomatal closure, and reduced photosynthesis, resulting in the onset of stress responses (Hillel, 2004).

To calculate root water uptake and transpiration, most models account for the quantity of water physically available in the soil and the capacity of plants to extract this water, as determined by root system characteristics, such as root density (Brisson et al., 2009). Root density functions vary with depth and depend on vegetation type, stage of growth, etc. (Jackson et al., 1996). In Richards' equation models, the root system is typically represented as a spatially distributed sink term extending to a specific rooting depth (often linked to plant growth), with uptake governed by soil hydraulic conditions and vegetation properties, such as the root distribution, growth stage and physiological stress responses (Feddes et al., 1978; Jarvis, 1989). These factors are incorporated into U in Equation (4), leading to feedback loops between soil moisture conditions and root water uptake that require careful representation within models (Feddes et al., 2001). Lu et al. (2020) identify the many ways that roots influence the soil hydraulic properties, although none of the models that were assessed appeared to consider these phenomena. Tipping-bucket models distribute root water uptake across soil layers (buckets) according to root density, with higher extraction near the surface where root density is greatest (Neitsch et al., 2011). The factors considered in the root water uptake functions of 16 different models are presented in Table 3.

Table 3. Root water uptake functions and associated parameters in soil-crop hydrology models. Models 1-14 are primarily 1D soil-crop hydrology models, while models 15 and 16 (marked with "*") are semi-distributed catchment models. Where there was no evidence in the user manuals or the key papers that a given approach was adopted in the model, "-" is used.

No.	Model	Root water uptake function**	Factors considered in estimating the rate of water uptake by plants**						
			Potential transpiration	Root depth	Root length density	Root resistance term	Unsaturated soil hydraulic conductivity	Soil water content	Soil water potential
1	APSIM	Passioura (1983) and Monteith (1986)[1]	-	-	-	-	-	-	-
2	LEACHM	Nimah and Hanks (1973)[2]	-	Yes[2]	Yes[2]	Yes[2]	Yes[2]	-	Yes[2]
3	SPLASH	By a *crop water stress factor*[#3]	-	-	-	-	-	Yes[3]	-
4	SWAP	Feddes (1978)[4]	Yes[5]	Yes[5]	Yes[5]	-	-	Yes[5]	Yes[5]
5	CropSyst	Campbell (1985)[6]	Yes (as a function of the LAI)[6]	Yes[6]	Yes[6]	-	Yes[6]	Yes[6]	Yes[6]

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6	HYDRUS-1D	Feddes (1978)[7]	Yes[7]	Yes[7]	Yes[7]	-	-	Yes[7]	Yes[7]
7	SALUS	Ritchie (1998)[8]	-	Yes[8]	-	-	-	Yes[8]	-
8	RZWQM2	Nimah and Hanks (1973)[9]	-	Yes[9]	Yes[9]	Yes[9]	Yes[9]	-	Yes[9]
9	DSSAT	Ritchie (1998)[10]	Yes (as a function of the LAI)[10]	Yes[10]	Yes[10]	-	Yes[10]	Yes[10]	Yes[10]
10	STICS	-	Yes[11]	-	Yes[11]	-	-	Yes[11]	-
11	AquaCrop	Feddes (1978)[12]	Yes[12]	Yes[12]	-	-	-	Yes[12]	-
12	EPIC	-	Yes[13]	Yes[13]	-	-	-	Yes[13]	-
13	WAVES		Yes[14]	Yes[14]	Yes[14]			Yes[14]	Yes[14]
12	Daisy	-	-	Yes[15]	Yes[15]	Yes[15]	-	Yes[15]	Yes[15]

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13	MIKE SHE*	Kristensen and Jensen (1975)[16]	-	Yes[16]	Yes[16]	-	-	-	-
14	SWAT*	-	Yes[17]	Yes[17]	Yes (water-use distribution parameter[#])[17]	-	-	Yes[17]	-

[⁂]*Crop water stress factor* in SPLASH quantifies the reduction in crop water uptake due to soil moisture depletion, with a value of 1 at/above FC and 0 at PWP.

[#]Water-use distribution parameter in SWAT determines the vertical distribution of root water uptake; a higher value means more water is taken.

[⁂]Sources: (1) Holzworth et al., 2014; (2) Hutson, 2003; (3) Arunakumaren, 1997; (4) Anuraga et al., 2006; (5) van Dam et al., 1997; (6) Stöckle and Nelson, 2003; (7) Šimůnek and Hopmans, 2009; (8) Basso and Ritchie, 2015; (9) Ma et al., 2012; (10) Jones et al., 2003; (11) Brisson et al., 2009; (12) Steduto et al., 2009; (13) Sharpley and Williams, 1990; (14) Zhang and Dawes, 1998; (15) Abrahamsen and Hansen, 2000; (16) DHI, 2024; (17) Neitsch et al., 2011.

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As shown in Table 3, the parameterisation of root systems varies between the models. Among the reviewed models, 11 of the 16 models can account for differences in both the penetration depths of roots and the density of roots, where the latter is simulated using a root length density term, which reflects the total root length per unit volume of soil (Wang and Smith, 2004). Three out of the 16 models also explicitly simulate root resistance, which is the resistance to water flow within the root system, reflecting frictional losses and hydraulic limitations within the root pathway (Nimah and Hanks, 1973). APSIM applies an alternative approach to root water uptake, adopting a layer-specific extraction coefficient ("kl") to define the maximum fraction of plant-available water that can be extracted (per unit time) by vegetation from each soil layer (Holzworth et al., 2014).

Under certain situations, plant roots can extract water from the saturated zone, although it is commonly presumed that root water uptake primarily occurs within the unsaturated zone (Jarvis, 2011). For example, MIKE SHE distributes evapotranspiration between the unsaturated and saturated zone cells under two conditions: (a) when the water table rises into the root zone, and (b) when phreatophytic vegetation extends roots into the capillary fringe and/or saturated zone (DHI, 2024).

3.4 Crop growth

Crop growth is an important factor that typically needs to be accounted for in soil-crop hydrology models of agricultural settings because dynamic interactions between soil water movement and plant water use vary over timescales of interest, particularly in irrigated systems (Feddes et al., 1988). Crop growth may be prespecified (i.e., independent of soil conditions) or adaptive, where the latter may apply simple empirical relationships or complex functions that represent plant physiological processes. Crop growth is reduced and senescence (i.e., decline of physiologically active biomass and leaf area) is accelerated under water and/or salinity stress (van Dam et al., 1997) and nitrogen limitation (Wang et al., 2002). Nitrogen availability is controlled by soil organic matter, carbon-to-nitrogen ratios, fertiliser management, crop demand, and environmental controls on microbial activity (Silva et al., 2023; van Ittersum et al., 2003; Yin et al., 2001). In crop-growth models, these controls are represented to varying degrees. More detailed crop-growth models simulate nitrogen pools, mineralisation-immobilisation processes, fertiliser additions, plant uptake and leaching (Brisson et al., 1998; Holzworth et al., 2014; Jones et al., 2003; Stöckle et al., 2003), whereas simpler approaches represent nutrient limitation using empirical stress factors (Raes et al., 2023). These differences are important because nitrogen stress can influence simulated crop growth, transpiration demand and root development, thereby affecting estimates of irrigation demand, drainage and recharge (van Ittersum et al., 2003; Wang et al., 2002). The main characteristics of the crop growth approaches used by different models are outlined in Table 4.

Table 4. Crop growth modelling approaches of soil-crop hydrology models. Models 1-14 are primarily 1D soil-crop hydrology models, while models 15 and 16 (marked with "*") are semi-distributed catchment models. Where there was no evidence in the user manuals or the key papers that a given approach was adopted in the model, "-" is used.

No.	Model	Crop growth components[*,**]					Stress factors on crop growth[*,**]		
		Biomass production	Canopy growth	Root growth	Phenological stage	Senescence / death	Water stress	Salinity stress	Nitrogen limitation
1	APSIM	Yes[1]	Yes[1]	Yes[1]	Yes[1]	Yes[1]	Yes[1]	Yes[2]	Yes[1]
2	LEACHM	No[3]	Yes (via the crop cover fraction)[3]	Yes[3]	-	-	Yes[3]	Yes[3]	-
3	SPLASH	No[4]	-	Yes[4]	-	-	No[4]	No[4]	-
4	SWAP-WOFOST	Yes[5]	Yes[5]	Yes[5]	Yes[5]	Yes[5]	Yes[5]	Yes[5]	Yes[6]
5	CropSyst	Yes[7]	Yes[7]	Yes[7]	Yes[7]	Yes[7]	Yes[7]	Yes[7]	Yes[7]
6	HYDRUS-1D	No[8]	-	Yes[8]	-	-	No[8#]	No[8#]	-

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7	SALUS	Yes[9]	Yes[9]	Yes[9]	Yes[9]	-	Yes[9]	-	Yes[9]
8	RZWQM2	Yes[10]	Yes[10]	-	Yes[10]	-	Yes[10]	-	Yes[10]
9	DSSAT	Yes[11]	Yes[11]	Yes[11]	Yes[11]	Yes[11]	Yes[11]	-	Yes[11]
10	STICS	Yes[12]	Yes[12]	Yes[12]	Yes[12]	Yes[12]	Yes[12]	-	Yes[12]
11	AquaCrop	Yes[13]	Yes[13]	Yes[13]	Yes[13]	Yes[13]	Yes[13]	Yes[14]	Yes (via soil fertility stress coefficient[#])[15]
12	Daisy	Yes[16]	Yes[16]	Yes[16]	Yes[16]	Yes[16]	Yes[16]	-	Yes[16]
13	EPIC	Yes[17]	Yes[17]	Yes[17]	Yes[17]	Yes[17]	Yes[17]	Yes[18]	Yes[17]
14	WAVES	Yes[19]	Yes[19]	Yes[19]	-	-	Yes[19]	Yes[19]	Yes[19]
15	MIKE SHE*	No[20]	Yes (via the LAI)[20]	Yes[20]	-	-	No[20]	-	-
16	SWAT*	Yes[21]	Yes (via the canopy height)[21]	Yes[21]	Yes[21]	-	Yes[21]	-	Yes[21]

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[⁂]The soil fertility stress coefficient represents the reduction in crop growth due to nutrient limitations. It is user-specified or calibrated based on field evidence of the soil fertility levels and corresponding crop growth response.

[⁂⁂]Sources: (1) Holzworth et al., 2014; (2) Sarkar et al., 2022; (3) Hutson, 2003; (4) Arunakumaren, 1997; (5) van Dam et al., 1997; (6) Groenendijk et al., 2016; (7) Stöckle et al., 2003; (8) Šimůnek et al., 2023; (9) Basso and Ritchie, 2015; (10) Ma et al., 2012; (11) Jones et al., 2003; (12) Brisson et al., 1998; (13) Raes et al., 2009; (14) FAO, 2012; (15) Raes et al., 2023; (16) Hansen, 2002; (17) Sharpley and Williams, 1990; (18) Yang et al., 2019; (19) Zhang and Dawes, 1998; (20) DHI, 2024; (21) Neitsch et al., 2011.

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According to Table 4, 12 of the 16 models (APSIM, SWAP-WOFOST, CropSyst, SALUS, SWAT, DSSAT, STICS, AquaCrop, Daisy, RZWQM2, EPIC and WAVES) include dynamic crop growth functions that simulate biomass production using representations of plant physiological processes. One or more of the following drivers are included in modelling biomass production in crop growth models: carbon assimilation, radiation use efficiency and/or crop transpiration (Todorovic, 2009; Uniyal and Dietrich, 2021). The produced biomass is partitioned into different organs (e.g., leaves, stems and roots) according to phenological stage (e.g., germination, flowering and maturity) within APSIM, SWAP-WOFOST, CropSyst, SALUS, DSSAT, STICS, Daisy RZWQM2 and EPIC, requiring detailed knowledge of crop growth dynamics (Wang et al., 2002). AquaCrop and SWAT also simulate biomass production and root/canopy development, although their biomass-allocation approaches are more simplified than those in more detailed crop-growth models (Neitsch et al., 2011; Raes et al., 2009).

The models in Table 4 that do not estimate biomass accumulation (e.g., LEACHM, SPLASH, HYDRUS-1D and MIKE SHE) include simplified vegetation representations based mainly on empirical functions, as their main focus is on soil hydrological processes. For example, LEACHM (Hutson, 2003) and MIKE SHE (DHI, 2024) include plant cover and root growth functions, while SPLASH (Arunakumaren, 1997) and HYDRUS-1D (Šimůnek et al., 2023) incorporate a simplified representation of vegetation dynamics, primarily through root depth dynamics. LEACHM (Hutson, 2003) and HYDRUS-1D (Šimůnek et al., 2023) also allow integration with external crop growth models to complement their soil water modelling capabilities. For example, the HYDRUS-1D model was integrated with WOFOST by Peña-Haro et al. (2012) to add a crop growth component.

According to Table 4, 13 of the 16 models include water stress as a factor affecting crop growth or crop-related vegetation development, while 7 models also incorporate salinity stress. HYDRUS-1D, SPLASH and MIKE SHE represent water stress through soil water processes that affect root water uptake, but none of these models simulates crop growth or yield responses to water stress. Models that simulate salinity stress on crop growth include some representation of solute transport within the soil, with varying levels of complexity. Simulating salinity stress also requires representation of solute inputs and transport processes within the soil, and in some models, relevant geochemical reactions (Feyen et al., 1998; Hillel, 2004; Leij and van Genuchten, 1999).

As shown in Table 4, all 12 models that simulate biomass accumulation incorporate nitrogen limitation as a factor influencing growth. Most models that simulate nitrogen limitation in crop growth represent nitrogen dynamics through relatively detailed nitrogen-cycling routines, including inorganic and organic nitrogen pools, mineralisation-immobilisation processes, plant uptake and leaching. Alternatively, AquaCrop represents nitrogen limitation using a soil fertility stress coefficient in the crop growth function without simulating the nitrogen cycle (Raes et al., 2023). Although HYDRUS-1D does not explicitly simulate nitrogen limitation effects on crop growth, it can represent nitrogen transport in the soil profile through its solute-transport capabilities (Šimůnek et al., 2023). Additionally, LEACHM has a separate version,

LEACHN, which extends the model to include detailed nitrogen cycling and crop uptake (Hutson, 2003).

3.5 Irrigation scheduling

In addition to meeting plant water requirements, irrigation also assists in managing soil salinity (Wichelns and Qadir, 2015). Thus, the irrigation water requirement is defined as that needed, in addition to precipitation, to produce the desired crop yield and to maintain an acceptable salt balance in the root zone (NRCS, 1993). Both the rate and timing of irrigation are important in the modelling of irrigated agricultural systems. Irrigation timing is determined by several factors, including: (a) plant responses to water deficit (e.g., reduction of transpiration due to inadequate available soil water (Blum, 2011)), (b) soil water content (Jones, 2004), (c) farmers' decisions (Jones, 2004), (d) rainfall variability (Nie et al., 2020), (e) soil salinity constraints (Dong et al., 2022; Wichelns and Qadir, 2015), and/or (f) limits to irrigation amounts imposed by water availability and allocation rules (Anthony and Birendra, 2018).

The amount of irrigation applied in scheduling methods is determined in several alternative ways. These include: (a) the amount of water required to refill the root zone, either up to FC or up to a user-defined fraction of the plant-available water deficit below FC (Champaneri et al., 2024), or (b) a predefined fixed irrigation depth applied uniformly regardless of the soil water status (Itier et al., 1996). Allen et al. (1998) and Jones (2004) emphasised that irrigation scheduling should ensure that the soil can provide water to plant roots at a rate that meets transpiration demand while minimising losses from evaporation and excessive seepage past the root zone. Li et al. (2020), Pan et al. (2020), and Yang-Ren et al. (2007) additionally consider soil salinity in optimising irrigation scheduling. The irrigation scheduling options available in different models are presented in Table 5.

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Table 5. Irrigation scheduling options available in soil-crop hydrology models. Models 1-14 are primarily 1D soil-crop hydrology models, while models 15 and 16 (marked with "*") are semi-distributed catchment models. Where there was no evidence in the user manuals or key papers that a given approach is available in the model, "-" is used.

No.	Model	Irrigation timing*,**			Amount of irrigation*,**	
		Based on rainfall and/or evapotranspiration criteria	Based on soil water content (θ) or the soil water potential (Ψ)	Based on fixed scheduling (fixed interval and/or fixed dates)	Based on soil water content (θ)	Based on prescribed irrigation depth, rate or duration
1	APSIM	Yes (based on rainfall)[1]	Yes (based on user-defined allowable soil water deficit[#])[1]	Yes (irrigate on a specified event (e.g., sowing), on a specified date, or at the start of the week)[1]	Yes (refill the soil profile up to a user-defined θ)[1]	Yes[1]
2	LEACHM	Yes (based on rainfall)[2]	Yes (based on user-defined Ψ)[2]	Yes[2]	Yes (refill the soil profile up to FC)[2]	Yes[2]
3	SPLASH	-	Yes (based on user-defined allowable soil water deficit[#])[3]	Yes[3]	Yes (refill the soil profile up to a user-defined θ or FC)[3]	Yes[3]

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4	SWAP	Yes (based on the ratio of actual transpiration to potential transpiration)[4]	Yes (based on user-defined allowable soil water deficit※, or Ψ)[4]	-	Yes (refill the soil profile up to FC)[4]	Yes[4]
5	CropSyst	-	Yes[5]	Yes[5]	Yes (refill the soil profile up to a user-defined θ)[5]	Yes[5]
6	HYDRUS-1D	-	Yes (based on user-defined Ψ)[6]	Yes[6]	-	Yes (by user-defined irrigation rate, duration and timing)[6]
7	SALUS	-	-	Yes (on user-defined date or days after planting)[7]	Yes (refill the soil profile up to a user-defined θ)[7]	Yes[7]
8	RZWQM2	Yes (based on evapotranspiration requirement)[8]	Yes[8]	Yes[8]	Yes (refill the root zone)[9]	Yes[8]

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9	DSSAT	-	Yes[10]	Yes[10]	Yes (refill the soil profile up to a user-defined θ)[10]	Yes[10]
10	STICS	Yes (based on the ratio of actual transpiration to potential transpiration)[11]	-	Yes[11]	Yes (refill the root zone up to FC)[11]	Yes[11]
11	AquaCrop	-	Yes (based on user-defined allowable water deficit[⁑])[12]	Yes[12]	Yes (refill the root zone up to FC)[12]	Yes[12]
12	Daisy	-	Yes (based on user-defined Ψ)[13]	Yes[13]	-	Yes (user-defined amount)[13]
13	EPIC	-	Yes[14]	Yes[14]	Yes[14]	Yes[14]
14	WAVES	-	-	Yes[15]	-	Yes[15]
13	MIKE SHE*	Yes (based on the ratio of actual evapotranspiration to potential crop evapotranspiration)[16]	Yes (based on user-defined allowable soil water deficit[⁑])[16]	Yes[16]	Yes (refill the soil profile up to a user-defined θ)[16]	Yes[16]

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14	SWAT*	Yes (based on the ratio of actual transpiration to potential transpiration)[17]	Yes[17]	Yes[17]	-	Yes (by user-defined maximum irrigation amount)[17]

[⁑]Allowable soil water deficit is the fraction of plant-available water the crop can extract before water stress occurs.

[⁂]Sources: (1) Holzworth et al., 2014; (2) Hutson, 2003; (3) Arunakumaren, 1997; (4) van Dam et al., 1997; (5) Stöckle and Nelson, 2003; (6) Šimůnek et al., 2013; (7) Basso, 2023; (8) Ma et al., 2011; (9) Hanson et al., 1998; (10) Jones et al., 2003; (11) Brisson et al., 2009; (12) Raes et al., 2023; (13) Hansen, 2002; (14) Sharpley and Williams, 1990; (15) Zhang and Dawes, 1998; (16) DHI, 2024; (17) Neitsch et al., 2011.

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Irrigation simulation models typically provide two main scheduling approaches: (a) irrigation applied according to user-defined schedules or intervals (e.g., calendar dates, days after planting or fixed irrigation schedules), and (b) automatic irrigation initiated when a user-defined soil moisture threshold (e.g., 50% of the plant-available water (Raes et al., 2023)) is reached (Uniyal and Dietrich, 2021). As summarised in Table 5, 13 of the 16 reviewed models represent irrigation triggering using soil water content-based criteria. Alternatively, models such as SWAP, SWAT, STICS and MIKE SHE consider actual crop evapotranspiration relative to potential evapotranspiration as an option for irrigation triggering. Once the ratio of actual to potential evapotranspiration falls below a user-defined threshold, these models trigger irrigation according to the specified management rules.

Of the 16 reviewed models, 13 determine irrigation amounts based on soil water content. Models such as LEACHM, SWAP and STICS automatically refill the soil profile up to FC, whereas models like APSIM, SPLASH, CropSyst, SALUS, DSSAT, EPIC and MIKE SHE allow irrigation up to a user-defined soil water content, enabling the simulation of deficit irrigation by refilling the soil profile to less than FC. This irrigation approach can also be used to manage soil salinity by applying additional water specifically for salt leaching whenever necessary (van Dam et al., 1997). In SPLASH, salinity management is achieved through a leaching factor, which permits overwatering to enhance deep drainage and reduce soil salt content. The leaching factor is defined as the ratio of the irrigation demand required to achieve a predefined leakage to the irrigation demand needed to meet the allowable soil water deficit (Arunakumaren, 1997).

3.6 Boundary conditions

Two boundary conditions are required for soil-crop hydrology models: (a) the upper boundary representing land-surface conditions, and (b) the lower boundary at the base of the simulated profile (Maskey, 2022). These boundary conditions strongly influence simulated soil-water dynamics and the estimation of drainage and recharge. In particular, the upper boundary governs the partitioning of incident water (precipitation and irrigation) into infiltration, runoff, surface evaporation and, where allowable, temporary surface ponding (Aravena and Dussaillant, 2009). This is especially important in 1D models because the upper boundary determines the flux of water entering the profile.

In tipping-bucket models, the upper boundary is usually represented as a net flux at the soil surface, corresponding to the balance between incident water and surface losses (Romano et al., 2011). In Richards' equation models, the upper boundary is typically represented using a surface node and may be treated either as a specified-flux boundary or, when surface saturation or ponding occur, as a specified-head boundary, with switching between these conditions depending on whether ponded water is present (Hutson, 2003; Šimůnek et al., 2023; van Dam et al., 1997). Under a specified-flux boundary, infiltration is limited by the infiltrability of the soil (Hillel, 2004), which depends on the initial θ, Ψ and soil hydraulic properties such as texture and structure (Hillel, 2004). Upward discharge at the land surface may also occur

through evaporation, including vapour diffusion driven by gradients in vapour pressure and temperature between the soil and the atmosphere (Han and Zhou, 2013).

In models that explicitly allow surface ponding (e.g., LEACHM, HYDRUS-1D and SWAP), once ponding begins the upper boundary is governed by a pressure head equal to the depth of ponded water (Hutson, 2003; Maskey, 2022; van Dam, 1997). Ponded water is then tracked through a surface water balance to maintain mass conservation (Hutson, 2003; Šimůnek et al., 2016; van Dam, 1997). Ponding may continue until a surface storage threshold is exceeded, commonly referred to as depression storage or rill storage (Abedini et al., 2006; Dunne and Leopold, 1978), after which excess water is routed as surface runoff while infiltration remains governed by the soil hydraulic properties (van Dam, 1997). This treatment is commonly used in soil water models that solve Richards' equation.

The lower boundary conditions may also be important, particularly where the water table is shallow or where upward flow from groundwater can influence the root-zone water balance (Feddes et al., 2001; He et al., 2021; Šimůnek et al., 2016). Carrera-Hernández et al. (2012) recommend using a fixed-head lower boundary (Dirichlet-type condition) for unsaturated flow modelling in sub-humid and semi-arid regions, noting that a unit-gradient boundary cannot represent upward flux during dry periods. Ordens et al. (2014) similarly applied a fixed-head lower boundary and found that the water-table depth strongly affected the timing and seasonality of recharge in the semi-arid Uley South Basin, South Australia. More generally, a fixed-head lower boundary is appropriate where the water table is shallow enough to influence soil-moisture dynamics. Where the water table is sufficiently deep to have minimal influence on the simulated profile, free drainage is commonly used as an approximation of gravity-driven downward flow (Feddes et al., 1988). This corresponds to a Neumann-type lower boundary in which the hydraulic gradient is prescribed, typically as a unit gradient, so that water drains freely from the profile under gravity. The choice of lower boundary condition therefore directly affects simulated drainage below the root zone and, hence, groundwater recharge estimates (He et al., 2021; Šimůnek et al., 2016).

When horizontal subsurface drainage (e.g., tile drains) is considered, some Richards' equation models incorporate the Hooghoudt (1940) (Appendix B, Equation B1) drainage concept, including HYDRUS-1D (Šimůnek et al., 2013), LEACHM (Hutson, 2003), SWAP (van Dam et al., 1997), and RZWQM2 (Ma et al., 2012). The Hooghoudt concept is an analytical steady-state solution for groundwater flow to parallel drains, in which drainage flux is expressed as a function of water-table height above the drains, drain spacing, and soil hydraulic conductivity. In 1D soil models, it is typically implemented as a head-dependent (Cauchy-type) lower boundary condition that approximates lateral flow to drains within a vertically resolved framework (Feddes et al., 1988; Hooghoudt, 1940; Ritzema, 1994).

Tipping-bucket models usually represent tile drainage more empirically. For example, in SWAT, tile drainage occurs when the water table rises above the depth of the installed drains, and the drainage rate is calculated using the parameters: (a) drain depth, (b) the time required

669 to drain the soil to FC, and (c) a lag term representing the travel time through the tile drainage
670 system before discharging to the channel network (Appendix B, Equation B2) (Neitsch et al.,
671 2011).

672

673 **4 Coupled estimation of groundwater pumping and recharge: Modelling case studies**

674

675 This section reviews 13 case studies in which irrigation demand and/or groundwater extraction
676 scenarios were evaluated together with deep drainage or recharge using the models described
677 in Section 3. Table 6 summarises the model applications in the selected case studies.

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678 **Table 6.** Summary of modelling case studies where irrigation demand and/or pumping, and deep drainage or recharge, were estimated using soil-
679 crop hydrology models. Where no corresponding information was reported in the case study, "-" was used.

Case Study	Location of the study	Application classification	Water source[#]	Soil-crop hydrology model	Target predictions	Uses of model outputs in groundwater investigations	Datasets used in model calibration	Datasets used to validate simulation outputs
Arunakumaren et al. (1999)	Burdekin River Irrigation Area, North Queensland, Australia	Planning/decision-support	CU	SPLASH	Deep drainage and irrigation demand	The SPLASH outputs were applied (using STRESGEN[*]) as inputs to MODFLOW for simulating the groundwater level changes	Known irrigation water use (for SPLASH)	-
Chen et al. (2018)	Northern High Plains of Texas, USA	Decision-support/analysis of existing system	GW	SWAT	Irrigation, evapo-transpiration, deep drainage and crop yield	The difference between irrigation and deep drainage was considered as net groundwater usage,	LAI, evapo-transpiration, and crop yield data	LAI, evapo-transpiration, and crop yield data

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						assuming no net lateral subsurface flows		
Deines et al. (2021)	Northwest Kansas, USA	Analysis of existing system	GW	SALUS	Crop yield, evapo-transpiration, irrigation demand and deep percolation	-	Observed annual crop yield, metered pumping data, and evapo-transpiration derived from MODIS satellite data	-
Dourte et al. (2013)	Wargal watershed, Andhra Pradesh, India	Analysis of existing system	GW	SWAT	Deep drainage, irrigation demand, and runoff	The difference between irrigation and deep drainage was considered as net groundwater usage,	Reservoir storage volume time series	Estimated groundwater recharge derived from groundwater level measurements in monitoring wells using the water table fluctuation method (WTFM).

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						assuming no net lateral subsurface flows		
Hu et al. (2010)	Shijiazhuang Irrigation District, North China Plain	Planning/analysis of an existing system	GW	DSSAT and SWAT	Crop water use and deep drainage in DSSAT, and deep drainage in SWAT	DSSAT- and SWAT-estimated deep drainage were used as inputs to MODFLOW for simulating groundwater level changes	Field-measured specific leaf area (DSSAT), and runoff, agro-climatic and hydrological data (SWAT)	Field-measured groundwater level data
Hou et al. (2025)	Mu Us Sandy Land, Northwest China	Analysis of existing system	GW	HYDRUS-1D	Soil water content, actual evapo-transpiration, transpiration, and bottom flux (capillary Rise/deep drainage)	-	Laboratory-analysed soil texture and measured soil water content data	Measured soil water content, evapotranspiration and stable isotope analysis

Journal Pre-proofs

Kaur et al. (2014)	Ludhiana district, Punjab, India	Analysis of existing system	CU	CropSyst coupled with GIS	Irrigation demand, evapo-transpiration, soil moisture storage and deep drainage	Groundwater level changes were calculated by dividing the net recharge by the specific yield	-	Field-measured groundwater level data
Liu et al. (2018)	Zhangye basin, middle oasis of Heihe River basin, Northwest China	Analysis of existing system	CU	SWAP-EPIC	Deep drainage	SWAP-EPIC model-derived recharge coefficients were applied as direct inputs to a MODFLOW model to simulate groundwater level dynamics	Field-measured soil hydraulic properties and crop data (LAI, plant height and yield)	Remote sensing-based evapotranspiration data and groundwater level data
McMahon et al. (2012)	Burdekin River Irrigation Area, North	Decision-support	CU	APSIM	Deep drainage, nitrogen leaching, chloride leaching and	APSIM outputs were applied (using STRESGEN*) as inputs to MODFLOW	-	Groundwater level data

Journal Pre-proofs

	Queensland, Australia				irrigation demand	for simulating the groundwater level changes		
Morgan et al. (2015)	Lower Limestone Coast, South East of South Australia	Decision-support	GW	LEACHM	Deep drainage and irrigation demand	LEACHM outputs were used as inputs to MODFLOW for simulating the groundwater level changes	-	MODIS-derived evapotranspiration datasets and metered groundwater pumping data
Pulido-Velazquez et al. (2015)	Mancha Oriental system, Spain	Planning/analysis of existing system	GW	SWAT	Deep drainage and irrigation demand and nitrate loads	SWAT-estimated pumping and recharge rates used as inputs to MODFLOW for groundwater level simulation	Literature-reported recharge data, crop survey information, and experimental crop yield and irrigation data	Field-measured groundwater level data

Journal Pre-proofs

Sun et al. (2015)	Luancheng Experimental Station, North China Plain	Analysis of existing system	GW	APSIM	Crop yield, irrigation demand and deep drainage	Groundwater level changes were calculated by dividing the net recharge (difference between deep drainage and pumping) by the specific yield	Observed flowering and maturity dates, and grain yield of wheat and maize	Field-measured grain yield, experimental evapotranspiration and field-measured soil water storage
Xiang et al. (2020)	Finney County, Kansas, USA	Analysis of existing system	GW	DSSAT	Crop yield, applied irrigation demand (pumping rates) and deep percolation (recharge rates)	DSSAT-estimated pumping and recharge rates were used as inputs to MODFLOW for simulating groundwater levels	-	Field-measured groundwater level data and crop yield

[*]STRESGEN is a Fortran program that reads the data from GIS-created tables and the output from SPLASH and APSIM and combines these data to produce the files required by the packages in MODFLOW that simulate well pumping and recharge.

[#]GW = groundwater only; CU = conjunctive use of groundwater and surface water.

Journal Pre-proofs

The 13 case studies summarised in Table 6 form the basis for the discussion in Sections 4.1, 4.2 and 4.3. The case studies were classified according to their primary model application across planning, decision-support, and assessment stages, as well as their representation of irrigation water sources (groundwater only or conjunctive groundwater-surface water use). Section 4.1 examines how soil-crop hydrology models allocate irrigation demand between groundwater and surface-water sources, or otherwise prescribe pumping where source partitioning is not simulated explicitly. Section 4.2 focuses on approaches used to validate model outputs, and Section 4.3 discusses the application of soil-crop hydrology models in planning, decision-support, and assessment stages in irrigation and water management.

4.1 Assigning irrigation demand to groundwater and surface-water sources in reviewed case studies

Of the 13 case studies listed in Table 6, 9 considered groundwater as the only source of irrigation water supply (Chen et al., 2018: SWAT; Deines et al., 2021: SALUS; Dourte et al., 2014: SWAT; Hou et al., 2025: HYDRUS-1D; Hu et al., 2010: DSSAT; Morgan et al., 2015: LEACHM; Pulido-Velazquez et al., 2015: SWAT; Sun et al., 2015: APSIM; Xiang et al., 2020: DSSAT), whereas 4 involved conjunctive use of surface water and groundwater (Arunakumaren et al., 1999: SPLASH; Kaur et al., 2014: CropSyst; Liu et al., 2018: SWAP-EPIC; McMahon et al., 2012: APSIM). This distinction is important because, once irrigation demand has been estimated using a soil-crop hydrology model, conjunctive-use applications require an additional step in which the demand is assigned to groundwater and surface-water sources. In doing so, it is necessary to consider both the timing and spatial application of irrigation from groundwater and surface water sources, together with dependencies on climate conditions, crop cycles, water availability, and, in some settings, water costs (Pongkijvorasin and Roumasset, 2007). More broadly, the conjunctive-use literature shows that this assignment is commonly based on one or more of four approaches: (a) historical or prescribed-source partitioning, (b) priority-based allocation according to source availability, (c) calibration-constrained allocation using observed abstraction or delivery data, and (d) optimisation-based allocation designed to satisfy production, economic or environmental objectives (Evans and Evans, 2012; Foster and van Steenbergen, 2011; Foster et al., 2010; Pulido-Velázquez et al., 2006; Ross, 2018; Singh, 2014).

The simplest approach is to partition irrigation source proportions using historical abstraction records, delivery data, or survey information. Murphy et al. (2005) and McMahon et al. (2012) separated irrigation demand in the Pioneer Valley and Burdekin River irrigation areas, respectively, into groundwater and surface-water components using expected ratios of groundwater and surface-water supply. In these applications, fixed fractions of the simulated irrigation demand were assigned to groundwater within subregions of the study area, with the remainder supplied from surface water. The allocation fractions were derived from metered abstraction data where available, or from landholder survey information where direct measurements were unavailable (e.g., Arunakumaren et al., 2001). This type of prescribed allocation is straightforward to implement, but its reliability depends strongly on the quality and representativeness of the data used to define the partitioning of demand between sources.

A second approach is to allocate irrigation according to source availability, such that surface water is used first and groundwater supplies the residual demand. Kaur et al. (2014) adopted this type of priority-based allocation in the Ludhiana district in Punjab, India. In that study, block-wise canal water supplies were subtracted from the simulated irrigation requirement to estimate spatially variable groundwater demand. Canal water allocation for each block was derived from daily canal release records and spatially distributed according to the canal network length within each block using GIS-based overlay analysis. This type of allocation is consistent with broader conjunctive-use practice, where groundwater commonly acts as a buffer against shortfalls in surface water reliability or delivery timing (Evans and Evans, 2012; Foster and van Steenbergen, 2011; Foster et al., 2010).

Where data are available, irrigation source partitioning should ideally be constrained using observed pumping or delivery records. However, among the reviewed studies, only Arunakumaren et al. (1999) appears to have partitioned total irrigation demand between surface water and groundwater using both metered groundwater and metered surface-water data. In that study, the Burdekin River Irrigation Area was divided into groundwater management zones, and time-varying allocation factors were used to define the proportion of irrigation supplied from each source according to metered records. Arunakumaren et al. (1999) also explored optimisation of the groundwater-surface water split to identify conjunctive-use combinations that would mitigate environmental impacts such as rising water tables. The broader conjunctive-use literature includes other such examples, in which the partitioning of irrigation demand between groundwater and surface water is influenced by source reliability, delivery timing, pumping cost, water quality, and environmental constraints, and may be assigned explicitly within optimisation studies when sufficient supporting data are available (Evans and Evans, 2012; Pulido-Velázquez et al., 2006; Ross, 2018).

In addition to the selected case studies, Safavi and Esmikhani (2013) applied a simulation-optimisation framework to evaluate the conjunctive use of surface water and groundwater in the Zayandehrood River Basin, Iran. The model incorporated constraints on cumulative groundwater drawdown and irrigation system capacity. Their results demonstrated that the proposed approach could reduce irrigation water shortages through more efficient allocation of available resources. Similarly, Pulido-Velázquez et al. (2006) used a hydrologic-economic optimisation model to allocate water between surface and groundwater sources by maximising net economic surplus, integrating surface hydrology, transient groundwater flow, and stream-aquifer interactions within a nonlinear framework. Their findings highlighted the ability of optimisation-based approaches to derive economically efficient allocation strategies while providing estimates of marginal water values and system constraints. Singh (2014) reviewed conjunctive water use for sustainable irrigated agriculture, with particular emphasis on optimisation-based approaches, noting that while linear programming models are widely applied due to their simplicity, their limited capacity to represent nonlinear processes has led to the adoption of more advanced techniques, including nonlinear programming, dynamic programming, and stochastic optimisation.

Beyond the reviewed soil-crop hydrology models, integrated hydrological and water-management frameworks provide additional approaches for representing groundwater–surface water interactions and source allocation in conjunctive applications. For example, MODFLOW coupled with river/reservoir operations model MODSIM was used to evaluate demand-based conjunctive water use by linking groundwater dynamics with water availability and management decisions (Morway et al., 2016). Additionally, Kitlasten et al. (2021) applied an integrated operation-hydrology modelling framework developed by Morway et al. (2016), with the groundwater-surface water flow model GSFLOW (Markstrom et al., 2008), to dynamically simulate conjunctive surface-water and groundwater use under projected climate change scenarios. Although these integrated frameworks generally lack the detailed crop growth and soil-process representation of soil-crop hydrology models, they provide complementary approaches for representing groundwater–surface water interactions, water-source allocation, and management constraints.

Overall, the reviewed case studies indicate that assigning simulated irrigation demand to groundwater and surface water remains a major source of uncertainty in conjunctive-use applications. Most studies relied on prescribed source shares or surface-water availability constraints, whereas only one study appears to have directly constrained the source partitioning using metered data for both groundwater and surface water. In practice, conjunctive-use allocation is also influenced by water quality considerations, particularly where brackish groundwater requires mixing with surface water before irrigation, and suboptimal mixing ratios can lead to irrigation with water exceeding recommended salinity thresholds (Evans and Evans, 2012). This suggests that, in coupled estimation of recharge and pumping, uncertainty in source attribution may be as important as uncertainty in the estimation of total irrigation demand itself.

4.2 Calibration and validation of model simulations

To improve confidence in model behaviour and predictive reliability, soil-crop hydrology models are commonly calibrated against field-measured data to constrain crop and soil hydraulic properties (Jarvis et al., 2022) and to reduce discrepancies between simulated and observed system responses (Jarvis et al., 2022; Vereecken et al., 2016; Wallach et al., 2021; Wallach et al., 2024). Best practice is to calibrate against multiple observed variables simultaneously, such as phenology, LAI, soil water content, evapotranspiration, biomass and yield, because acceptable agreement for a single variable can mask compensating errors in parameter values or model structure (Wallach et al., 2021; Wallach et al., 2024). Management inputs such as irrigation timing and application depth should also be constrained, where possible, by field observations and knowledge of irrigation management practices rather than being treated as free calibration variables (Foster et al., 2017; Wallach et al., 2021; Wallach et al., 2024). For example, Liu et al. (2018) used measured irrigation records as inputs to the coupled SWAP-EPIC model, while field-measured soil hydraulic properties and crop variables, including LAI, plant height, and yield, were used for model calibration. Calibration against multiple observed variables and the use of field-constrained management inputs are particularly important in studies that aim to estimate recharge and pumping, because both outputs are sensitive not only to soil and crop parameters, but also to assumptions about irrigation management and landholder practices.

Table 6 summarises how the reviewed studies constrained, calibrated and/or evaluated soil-crop hydrology models using available field observations and other data sources, including crop measurements, metered pumping, remotely sensed evapotranspiration, soil water storage and groundwater levels. In several studies, calibration focused primarily on crop growth and irrigation-related parameters. For example, Sun et al. (2015) calibrated APSIM wheat and maize cultivar parameters by matching simulated and observed flowering and maturity dates, and then adjusted yield-related parameters to improve agreement with observed grain yield. Hu et al. (2010) improved the DSSAT representation of canopy development by replacing the default specific leaf area with a field-measured value after earlier work had identified substantial discrepancies between simulated and observed LAI. Arunakumaren et al. (1999) calibrated SPLASH using zone-specific irrigation use, crop factors and soil parameters, adjusting pan factors to match historical cane yield and allowable deficit parameters to reproduce monthly irrigation usage. Deines et al. (2021) first calibrated SALUS crop parameters against observed annual crop yields and subsequently adjusted irrigation management parameters to reproduce pumping records. Specifically, the authors iteratively evaluated 30 combinations of soil moisture thresholds and irrigation application depths to identify the parameter set that best matched recorded pumping volumes. Simulated irrigation volumes were further adjusted to improve agreement with observed pumping. This calibration approach reduced uncertainty associated with irrigation management assumptions and improved representation of water use, however, the extent to which calibration reduced uncertainty in simulated water-balance components was not quantified..

In addition to crop and irrigation parameters, some studies used broader indicators of model performance related to crop production and water use. Crop-growth variables such as yield, biomass, evapotranspiration and phenological development provide useful checks on whether the model is realistically representing plant water demand, soil-water redistribution and associated management responses (Yang et al., 2006). Among the reviewed studies, Sun et al. (2015) showed that APSIM successfully reproduced crop responses under different irrigation strategies and long-term climate variability in the North China Plain. However, model performance may decrease at the regional-scale applications due to limitations in representing spatial variability in climate, soil properties, and management practices. For example, Xiang et al. (2020) used crop yield outputs from the DSSAT-MODFLOW framework to evaluate trade-offs between groundwater conservation and agricultural productivity in Finney County, Kansas. The model reproduced crop yields with reasonable accuracy for some crops, including corn (13,677 kg ha^{-1} simulated vs. 13,980 kg ha^{-1} observed) and soybean (4,092 kg ha^{-1} simulated vs. 3,120 kg ha^{-1} observed). However, higher discrepancies were evident for sorghum and winter wheat, which the authors attributed to uncertainties in climate inputs and inconsistencies between the modelled irrigation domain and the county-level yield statistics used for evaluation.

Following calibration, several studies further evaluated soil-crop hydrology model outputs by testing whether the simulated recharge and pumping produced plausible groundwater responses. Seven of the 13 reviewed case studies applied soil-crop hydrology model outputs to MODFLOW to simulate groundwater-level changes and then compared the results with observed groundwater levels (Arunakumaren et al., 1999; Hu et al., 2010; Liu et al., 2018;

McMahon et al., 2012; Morgan et al., 2015; Pulido-Velazquez et al., 2015; Xiang et al., 2020). For example, Liu et al. (2018) calibrated the model against groundwater-level observations from 1991-2000 by adjusting key hydrogeological parameters, including hydraulic conductivity and specific yield. The calibrated model showed strong agreement between simulated and observed groundwater levels, with an RMSE of 2.6 m, NSE of 0.999, and R^2 of 0.998. Model performance remained strong during the independent validation period (2001-2010), with an RMSE of 3.9 m, NSE of 0.996, and R^2 of 0.998. Similarly, Morgan et al. (2015) and Kaur et al. (2014) reproduced regional groundwater-level trends with reasonable accuracy, despite local discrepancies in some areas. Morgan et al. (2015) reported that their transient regional model captured groundwater-level dynamics with an RMSE of 6.5 m and a scaled RMSE of 5.0%, although local discrepancies remained in some areas, with residual errors reaching up to 19 m. Kaur et al. (2014) also achieved accurate predictions of annual groundwater-level fluctuations, reporting an R^2 of 0.82 between observed and simulated groundwater levels.

Two other studies converted net recharge to estimates of water-table change using specific yield and compared the results with observed water-table fluctuations (Kaur et al., 2014; Sun et al., 2015). For example, Sun et al. (2015) compared annual water-table changes derived from APSIM-simulated net recharge and pumping with observed water-table fluctuations. Additionally, Hu et al. (2010) calibrated and validated the SWAT component using runoff, agro-climatic and hydrological data, thereby providing indirect constraints on the simulated mountain-front recharge inputs used in MODFLOW. These approaches provide useful indirect evaluation of recharge and pumping estimates, but do not uniquely validate those fluxes, because groundwater-level responses also depend on aquifer properties, boundary conditions, and the timing of recharge and discharge processes. As such, acceptable simulation of groundwater levels does not necessarily imply that recharge and pumping have been estimated accurately, and recharge uncertainty cannot be quantified directly from groundwater model performance alone (Knowling and Werner, 2017). This limitation is well recognised in the broader calibration literature, which emphasises the need for multiple and independent lines of evidence when evaluating model realism.

A small number of studies used data that more directly constrain irrigation demand and pumping. Morgan et al. (2015) compared modelled evapotranspiration with remotely sensed data, modelled irrigation with metered pumping data, and 1D LEACHM simulations with 2D simulations. These comparisons provided useful insights into the realism of simulated irrigation demand and led to adjustments in crop parameters within the 1D model. In particular, overestimated evapotranspiration in the drip-irrigation area was reduced by increasing the amount of surface mulch. Among the reviewed case studies, Morgan et al. (2015) appears to be the only study that explicitly used metered pumping data in validation, whereas Deines et al. (2021) used metered pumping data during calibration. Thus, direct evaluation of irrigation demand and pumping has rarely been undertaken, even where metered data were available.

Recharge estimates derived from soil-crop hydrology models have often been assessed using independent hydrogeological methods. For example, Dourte et al. (2013) validated simulated recharge against estimates obtained using the WTFM. Similarly, other studies have compared

907 recharge estimates derived from the WTFM with simulations from SWAT (Okrah et al., 2023),
908 LEACHM (Ordens et al., 2014), and HYDRUS-1D (Gumuła-Kawęcka et al., 2022; von
909 Freyberg et al., 2015). Anuraga et al. (2006) used a groundwater linear-store model driven by
910 SWAP recharge simulations and assessed the results against observed groundwater depths and
911 recharge estimates derived from water-table rise events. Hassen et al. (2021) applied the
912 chloride mass balance method and compared the resulting recharge estimates with those
913 simulated using HYDRUS-1D. These studies illustrate that independent hydrogeological
914 methods can provide valuable additional constraints on recharge estimates, especially where
915 groundwater-level comparisons alone are insufficient to distinguish between recharge
916 uncertainty and uncertainty in aquifer properties.

917

918 Overall, the reviewed case studies show that calibration and validation practices have not been
919 applied consistently in applications that estimate recharge and pumping. Most studies
920 calibrated at least some crop or irrigation parameters, but fewer undertook multi-variable
921 evaluation, and only a very small number directly assessed simulated pumping against metered
922 observations. Given the sensitivity of recharge and pumping estimates to irrigation
923 management assumptions, soil hydraulic properties and crop parameters, future work should
924 place greater emphasis on multi-variable calibration, explicit constraint of irrigation
925 management inputs, and the systematic use of metered pumping, remotely sensed
926 evapotranspiration, soil water observations and independent recharge estimates to improve
927 confidence in simulated recharge and pumping. This is particularly important where soil-crop
928 hydrology model outputs are subsequently used in groundwater assessments or water-
929 management decision making.

930

931 **4.3 Soil-crop hydrology model applications across planning, decision-support and**
932 **assessment stages**

933

934 The case studies listed in Table 6 can be evaluated according to their potential applications
935 during the planning stage (before irrigation), the decision-support stage (during irrigation), and
936 the assessment stage (after irrigation).

937

938 At the planning stage, decision-making relies on available information regarding water
939 resources, soil characteristics and crop management (Doorenbos and Pruitt, 1977). This stage
940 involves developing a tentative irrigation plan and refining it through staged analysis to identify
941 an appropriate management strategy (Doorenbos and Pruitt, 1977). At this stage, soil-crop
942 hydrology models are primarily used to generate long-term predictions for testing alternative
943 irrigation and water management strategies, often under conditions of limited field data, while
944 also providing information for estimating irrigation demand, groundwater recharge and
945 pumping. Hu et al. (2010) extended beyond simple recharge estimation by iteratively linking
946 DSSAT-, SWAT- and MODFLOW-derived fluxes to evaluate groundwater-equilibrium and
947 groundwater-recovery irrigation scenarios under reduced agricultural water use. Their
948 modelling framework was used to assess the effects of reduced crop water use on groundwater
949 decline and recovery, helping identify the irrigation reductions required to stop groundwater

drawdown and restore pre-development hydrologic conditions, while also evaluating trade-offs between water conservation and crop yield. Similarly, Arunakumaren et al. (1999) applied SPLASH to estimate irrigation demand, pumping requirements under conjunctive water use, and groundwater recharge across different land uses, thereby providing information to guide future water allocation and management decisions.

During the decision-support stage, soil-crop hydrology models are applied at finer spatial and temporal scales to support in-season management of irrigation, pumping and nutrient application, enabling farmers and water managers to optimise water use while minimising environmental risks such as soil salinity, nutrient leaching and groundwater over-extraction (Kisekka et al., 2017). These applications are more operational than planning-stage analyses and are intended to guide management decisions within irrigation seasons. Among the reviewed case studies, Arunakumaren et al. (1999), McMahon et al. (2012), and Morgan et al. (2015) developed location-specific modelling approaches that simulate soil water balance, crop growth and water uptake to support irrigation allocation and pumping decisions.

In the assessment stage, soil-crop hydrology models are used for retrospective analysis, focusing on broader, long-term patterns of water use, irrigation, and flow dynamics across regions. Kaur et al. (2014) assessed the impacts of different climatic factors (rainfall, evaporation, temperature), land use (crop types), irrigation strategies, and soil types on groundwater recharge using CropSyst and a water-balance approach. Sun et al. (2015) used the APSIM model to evaluate the effects of different irrigation strategies (full irrigation, critical stage irrigation, minimum irrigation, and rainfed) on crop yield and groundwater conservation. The results indicated that irrigation applied at critical growth stages of the wheat crop maximises irrigation water productivity (grain yield per unit depth of water applied). Xiang et al. (2020) applied a DSSAT-MODFLOW modelling framework to evaluate groundwater conservation and crop yield by assessing changes in irrigation technologies, crop selection, and climate change adaptation strategies. The model was applied in Finney County, Kansas, to perform a scenario analysis assessing the effects of pumping at current rates versus cessation of all pumping for the 2000-2007 period. Crop yield for both scenarios was also assessed. Deines et al. (2021) used the SALUS model to perform a post-hoc assessment of the first five-year cycle of the Sheridan-6 LEMA program. They quantified the irrigation efficiency, showing that while pumping decreased by 25%, only 22% of that reduction benefited the aquifer due to decreased irrigation return flow.

Model selection for specific applications requires careful consideration of the trade-off between process representation, calibration requirements, computational demand, and predictive performance. Increasing model complexity through more detailed representations of soil hydrology, such as physically based descriptions of water flow and solute transport, can improve the representation of key processes and enhance predictive capability under conditions where sufficient data are available. However, such improvements are often accompanied by increased parameter requirements, greater calibration complexity, and challenges associated with parameter uncertainty, particularly when models are applied across heterogeneous landscapes (Vereecken et al., 2016; Wallach et al., 2025; Zhang et al., 2020). Jarvis et al. (2022) highlighted this challenge by demonstrating that integrating detailed soil hydraulic processes

into crop modelling frameworks requires careful consideration of parameterisation and scaling approaches to ensure reliable predictions beyond the measurement scale. In contrast, models designed primarily for crop growth and management applications often adopt simplified soil water representations to reduce data requirements and computational burden, which can improve their applicability for large-scale assessments but may limit their ability to capture complex soil-water interactions. Therefore, increasing model complexity does not necessarily guarantee improved predictive performance; rather, model suitability depends on achieving an appropriate balance between process fidelity, calibration feasibility, computational efficiency, and the objectives of the application.

4.4 Remaining challenges and research needs

4.4.1 Incomplete representation of irrigation demand, losses and return flows

The reviewed model functions and case studies show that irrigation demand is not a single, directly observable quantity, but an inferred water-balance term that depends on how each model represents crop demand, irrigation scheduling, application losses, leaching requirements, runoff and drainage. Several processes that affect this calculation are represented inconsistently across models, including interception losses, leaching requirements for salinity control, and irrigation-induced runoff. Canopy interception is explicitly represented in some models as canopy water storage (i.e., the fraction of the precipitation or overhead irrigation that falls on the crop and is intercepted by and evaporated from the canopy (Stöckle et al., 1994)), while in some other models it is not represented as a separate canopy process and may instead by absorbed into empirical rainfall-runoff or initial-abstraction terms, such as those used in the CN approach (Neitsch et al., 2005). Similarly, leaching requirements for salinity control are explicitly considered in some models (e.g., Alsaeedi and Elprince, 1999; Shaw and Thorburn, 1985) by allowing salt accumulation and additional irrigation for leaching, whereas in others they are not directly simulated and must be accounted for externally (van Dam et al., 1997). Irrigation-induced runoff from applied water is often poorly represented across modelling frameworks, with some approaches incorporating irrigation into rainfall-runoff formulations (Holzworth et al., 2014; Jones et al., 2003).

Overall, these incomplete and inconsistent representations introduce uncertainty and potential bias that may result in either overestimation or underestimation of irrigation demand, depending on how different water-balance components are represented within the modelling framework. This is particularly important for groundwater studies because irrigation application, crop water use, deep drainage, return flow and aquifer recharge represent different components of the agricultural water balance, and errors in simulating one component can propagate into estimates of the others. Such inaccuracies not only affect on-farm water management decisions but can also translate into economic losses through reduced crop productivity, inefficient water use, or unnecessary energy costs associated with over-application (Qureshi et al., 2013a).

4.4.2 Limitations of 1D models for representing field and catchment processes

Natural soils and rainfall exhibit substantial spatial variability, and differences in infiltrability between regions can result in lateral redistribution of water that one-dimensional (1D) modelling approaches cannot capture (Hillel, 1998). As a result, the water balance components such as runoff and infiltration may be misrepresented, leading to biased estimates of water partitioning at the field scale. These limitations become particularly pronounced under non-uniform rainfall and irrigation conditions, such as furrow and border irrigation systems, where lateral flow processes and spatial variability in water application are not explicitly represented (Ebrahimian et al., 2012). This can lead to inaccurate simulation of soil moisture distribution and deep drainage losses.

In addition, 1D models typically assume vertically dominant water movement through the soil matrix, which restricts their ability to simulate preferential flow and rapid bypass pathways that can dominate infiltration and solute transport in structured soils (Jarvis, 2007; Šimůnek et al., 2002). Consequently, solute leaching and contaminant transport may be underestimated, overestimated or incorrectly timed, particularly under conditions where macropore flow is significant.

Representation of vertical soil-water-groundwater interactions is simplified in many 1D frameworks. Coupling between shallow groundwater, capillary rise, and root water uptake is highly idealised unless an appropriate lower boundary condition or groundwater couple is included, despite its strong influence on root-zone water availability and recharge dynamics, particularly in shallow water table environments (Bizhanimanzar et al., 2019). This simplification can result in errors in simulating plant water uptake, evapotranspiration, and groundwater contributions to the root zone. Recharge estimates are further affected by scaling challenges associated with spatial variability in climate, topography, soil, and geology (Arnold et al., 1998), which can propagate into regional-scale recharge estimates. To address spatial heterogeneity, several soil-crop hydrology models have been integrated with GIS platforms to varying degrees, enabling partial representation of spatial variability (e.g., LEACHM (Morgan et al., 2015), SWAP (Xu et al., 2012), CropSyst (Stöckle et al., 2014), HYDRUS-1D (Anlauf et al., 2018), SWAT (Winchell et al., 2010), DSSAT (Jones et al., 2003) and MIKE SHE (DHI, 2024)). However, GIS-based spatialisation does not by itself resolve lateral flow, run-on, preferential flow or feedbacks between irrigation, groundwater and crop water use; it mainly allows multiple 1D or semi-distributed simulations to be organised spatially.

To better capture lateral water redistribution, several enhanced modelling approaches incorporate simplified representations of lateral connectivity, such as run-on partitioning approaches that redistribute surface flow between adjacent units. For example, Arnold et al. (2010) incorporated the concept of run-on within the SWAT model by applying a partitioning coefficient that separates the portion of flow channelled off a landscape unit from the fraction

contributing to direct surface run-on to adjacent downslope units. Similarly, the SWAP model simulates run-on using a linearised approach analogous to that used for surface runoff, as summarized in Table 1 (van Dam et al., 1997).

Overall, the limitations of 1D models arise not simply from dimensionality, but from the difficulty of scaling profile-scale soil-water simulations to fields, farms and catchments were lateral redistribution, preferential flow, groundwater interactions and spatially variable irrigation practices may all affect drainage and recharge. These constraints highlight the need for improved process coupling and scalable modelling frameworks that more consistently integrate vertical and lateral hydrological dynamics within physically based modelling structures.

4.4.3 Weak validation of pumping, recharge and source partitioning

Validation of both pumping and recharge remains limited in many modelling studies. Pumping estimates are rarely evaluated against metered extraction data and are instead inferred from crop growth and irrigation-related parameters. Recharge is inherently difficult to measure directly (Ferre et al., 2007) and is therefore assessed using indirect approaches such as the WTFM (Gumuła-Kawęcka et al., 2022; Okrah et al., 2023; Ordens et al., 2014; von Freyberg et al., 2015) and chloride mass balance (Hassen et al., 2021). In many cases, soil-crop hydrology model outputs are further evaluated indirectly by assessing whether simulated recharge and pumping produce plausible groundwater responses, for example by coupling outputs to groundwater models such as MODFLOW to simulate water-level changes, or by converting net recharge to water-table fluctuations using specific yield and comparing these with observations.

These approaches are useful, but they do not uniquely validate recharge or pumping. Groundwater-level responses also depend on aquifer hydraulic properties, storage parameters, boundary conditions, groundwater abstractions, surface-water interactions and the timing of recharge and discharge. Therefore, agreement between simulated and observed groundwater levels can occur even where recharge and pumping are partly compensating or incorrectly partitioned. This limitation is especially important when soil-crop hydrology model outputs are used as groundwater model inputs, because calibration against heads alone may obscure uncertainty in the fluxes being transferred between models.

In addition, partitioning between groundwater and surface-water sources in conjunctive-use systems also remains weakly constrained. Most studies rely on prescribed source shares or simple surface-water availability constraints, with very limited use of metered data to constrain source attribution. However, conjunctive-use frameworks emphasise the dynamic and adaptive allocation of water sources, in which groundwater acts as a buffer to variability in surface-water supply and irrigation demand (Evans and Evans, 2012; Foster et al., 2010). Modelling

approaches that assume fixed or independently defined source contributions rarely capture the resulting temporal variations in groundwater pumping and surface-water use.

4.4.4 Next steps for model development and application

Future soil-crop hydrology models should prioritise validation and independent constraints of key water-balance fluxes, rather than relying only on internally consistent simulations. Greater emphasis is needed on the use of observational data, including metered pumping and irrigation delivery records, to constrain model estimates and reduce reliance on inferred parameters. The integration of remotely sensed evapotranspiration and soil moisture products also provides opportunities to better represent the timing and magnitude of irrigation across spatial scales (Zheng et al., 2024). In addition to physical processes, soil-crop hydrology models should more explicitly represent farmer behaviour, water availability constraints, and economic decision-making, recognising that irrigation demand is influenced by both biophysical and socio-economic factors (Qureshi et al., 2013b).

Salinity is another critical process that must be explicitly incorporated in soil-crop hydrology models, particularly in the estimation of irrigation demand where leaching requirements, shallow groundwater or saline irrigation water are important. Although some soil-crop hydrology models simulate hydraulic flow, solute transport, crop growth, and chemical processes within the soil profile (Kramer and Mau, 2023), they are generally not well suited for long-term simulations that account for seasonal variability and changing agricultural practices (Tunca et al., 2023). Future modelling should better capture the effects of irrigation water quality, salt accumulation and flushing, shallow water tables, drainage conditions, and, where relevant, sodicity and soil-structure change (Ayars et al., 2003). In many regions, salinity is not just a root-zone issue but a dynamic outcome of groundwater interaction, evapotranspiration, drainage efficiency, and water quality, so it should be treated as an integrated water-quality and hydrological process rather than as a separate leaching correction (Tunca et al., 2023).

Furthermore, there is a need for integrated modelling approaches in which soil-hydrology, crop, climate, agricultural management, groundwater and surface-water models are coupled rather than applied independently, to improve consistency across scales. Collectively, these advances will support more robust, data-constrained, and scalable modelling of irrigation demand and water balance processes (Wang et al., 2025). For conjunctive irrigation water management, such integration will be critical for developing realistic adaptation strategies under climate change, particularly where increasing frequency and severity of surface-water droughts coincide with rising water demand driven by higher temperatures (Foster et al., 2010). Further integration of optimisation frameworks with process-based soil-crop hydrology models would enable more effective allocation of water between sources by accounting for hydrological variability, water quality constraints, and management objectives, thereby improving the representation of conjunctive-use systems in irrigation-demand estimation (Singh, 2014).

5 Conclusions

The review demonstrates that soil-crop hydrology models differ substantially in their representation of key processes governing runoff generation, soil water movement, root water uptake, crop growth, irrigation scheduling, and boundary conditions, leading to variations in simulated irrigation demand and recharge under different hydrogeological and management settings. The reviewed models also differ substantially in how they represent salinity stress, nitrogen limitation, irrigation triggering, and groundwater interaction, all of which influence estimates of irrigation demand, drainage and recharge.

The key limitations identified across the literature are the inconsistent and often limited validation of groundwater pumping, recharge estimates, and conjunctive irrigation water allocation between surface and groundwater sources. Out of the reviewed case studies, only one study explicitly validated simulated pumping against metered data, while a limited number of studies used metered pumping data to inform the partitioning of irrigation demand between surface and groundwater sources. Most studies instead relied on indirect evaluation methods, such as comparing groundwater-level responses simulated using MODFLOW or water-balance approaches against observed groundwater fluctuations. While these approaches provide useful constraints, they do not uniquely validate recharge or pumping because groundwater responses are also influenced by aquifer properties, boundary conditions, and other hydrological stresses.

The review also highlights a conceptual limitation in many applications, where 1D representations are applied to systems in which lateral flow processes, preferential flow, salinity dynamics, and management feedback play an important role. This structural simplification can constrain the reliability of irrigation demand and recharge estimates, particularly in heterogeneous or intensively managed irrigated systems. More broadly, the review demonstrates that uncertainty in assigning irrigation demand between groundwater and surface-water sources may be as important as uncertainty in estimating total irrigation demand itself, particularly in conjunctive-use systems where water-source allocation varies through time according to water availability, delivery constraints and management decisions.

Future research should therefore focus on strengthening model validation against metered pumping data and irrigation delivery data, integrating remote sensing constraints and improving representation of conjunctive use allocation under surface-groundwater interactions. In addition, future modelling frameworks should incorporate more realistic representations of farmer decision-making, water availability constraints, and salinity-control irrigation practices to improve the robustness of irrigation demand and recharge estimation. Greater emphasis is also needed on multi-variable calibration and independent evaluation of recharge, irrigation demand and source partitioning using multiple lines of evidence, including groundwater responses, remotely sensed evapotranspiration, soil-water observations and hydrogeological recharge estimates.

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Author contributions: CRediT

E.M.G.P. Hemachandra: Methodology, Investigation, Visualization, Writing- Original draft preparation. **L.P. Reading:** Writing- Reviewing and Editing, Supervision. **A.D. Werner:** Conceptualization, Writing- Reviewing and Editing, Supervision, Project administration, Funding acquisition. **P. Egodawatta:** Writing - Review & Editing, Supervision.

Appendix A. List of abbreviations and variables

Table A1. Abbreviations used throughout the article

Abbreviation	Description
1D	One-dimensional
3D	Three-dimensional
APSIM	Agricultural Production Systems sIMulator
CN	Curve Number
CropSyst	Cropping Systems Simulation Model
DSSAT	Decision Support System for Agrotechnology Transfer model

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DUL	Drained Upper Limit
FC	Field capacity
LAI	Leaf Area Index
LEACHM	Leaching Estimation and Chemistry Model
NSE	Nash-Sutcliffe efficiency
PET	Potential Evapotranspiration
PWP	Permanent Wilting Point
R^2	Coefficient of Determination
RMSE	Root Mean Square Error
RZWQM2	Root Zone Water Quality Model
SALUS	System Approach to Land Use Sustainability
STICS	Multidisciplinary Simulator for Standard Crops
SWAP	Soil-Water-Atmosphere-Plant Model
SWAT	Soil and Water Assessment Tool

Table A2 Variables used throughout the article

Variable	Description	Units
E	Evaporation loss	LT^{-1}

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I	Infiltration	LT^{-1}
K_d	Vertical drainage-rate parameter related to vertical hydraulic conductivity (K_v [LT^{-1}]) in Darcy's law ($q = K_v(\Delta H/L)$ where q [LT^{-1}] is flux density and (ΔH/L) [-] is hydraulic gradient)	LT^{-1}
K_s	Saturated hydraulic conductivity	LT^{-1}
K_u	Unsaturated hydraulic conductivity	LT^{-1}
K_v	Vertical hydraulic conductivity	LT^{-1}
L	Leakage rate through the base of each soil layer (bucket)	LT^{-1}
q	Soil water flux	LT^{-1}
S	Degree of saturation	-
t	Time	T
T	Transpiration loss	LT^{-1}
U	Sink/source term representing water extraction and addition within the soil profile	T^{-1}
W	Water storage expressed as equivalent water depth	L
z	Vertical coordinate (positive upward)	L
Ψ	Soil water potential	L
θ	Soil water content	-
θ_r	Residual soil water content	-
θ_s	Saturated soil water content	-

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θ_u	Water content at which downward drainage is presumed to reduce to an insignificant rate, often presumed to be the field capacity (FC)	-
$\theta = \theta(\Psi)$	Water retention curve	-
$K_u = K_u(\theta)$ or $K_u(\Psi)$	Hydraulic conductivity function which is coupled to the moisture retention function via the water content or matric potential	LT^{-1}

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Appendix B. Tile drainage equations

B1. Hooghoudt equation for steady-state drainage

Drain discharge can be estimated by considering steady-state flow to parallel open drains with vertical walls and applying the Hooghoudt (1940) equation:

$$Q = \frac{4K_s\left(H^2 - D^2\right)}{L^2} \quad \text{(B1)}$$

where Q [LT^{-1}] is the drain discharge, H [L] is the height of the water table midway between the drains, D [L] is the elevation of the water level in the drain, and L [L] is the drain spacing (Ritzema, 1994).

B2. SWAT model tile drainage equation

In the SWAT model, tile drainage occurs when the water table rises above the elevation of the installed drains. The drain discharge on a given day is calculated as:

$$Q = \frac{h_w - h_d}{h_w}\left(d(\theta - \theta_u)\right)\left(1 - exp\left[\frac{-24}{t_d}\right]\right) \; h_w > h_d \quad \text{(B2)}$$

Where Q [L] is the drain discharge during the timestep, h_w [L] is the height of the water table above the impervious zone, h_d [L] is the height of the tile drain above the impervious zone, d [L] is the soil profile depth, θ [-] is the soil water content, θ_u [-] is the water content at which downward drainage is presumed to reduce to an insignificant rate, often presumed to be the field capacity (FC), and t_d [T] is the time required to drain the soil to FC (Neitsch et al., 2011).

Declaration of Generative AI and AI-assisted technologies in the manuscript preparation process.

During the preparation of this work, the authors used ChatGPT (GPT-5.5) to assist with language refinement, improving clarity, and checking readability. After using this tool, the authors thoroughly reviewed and edited the content as needed and take full responsibility for the accuracy and integrity of the published article.

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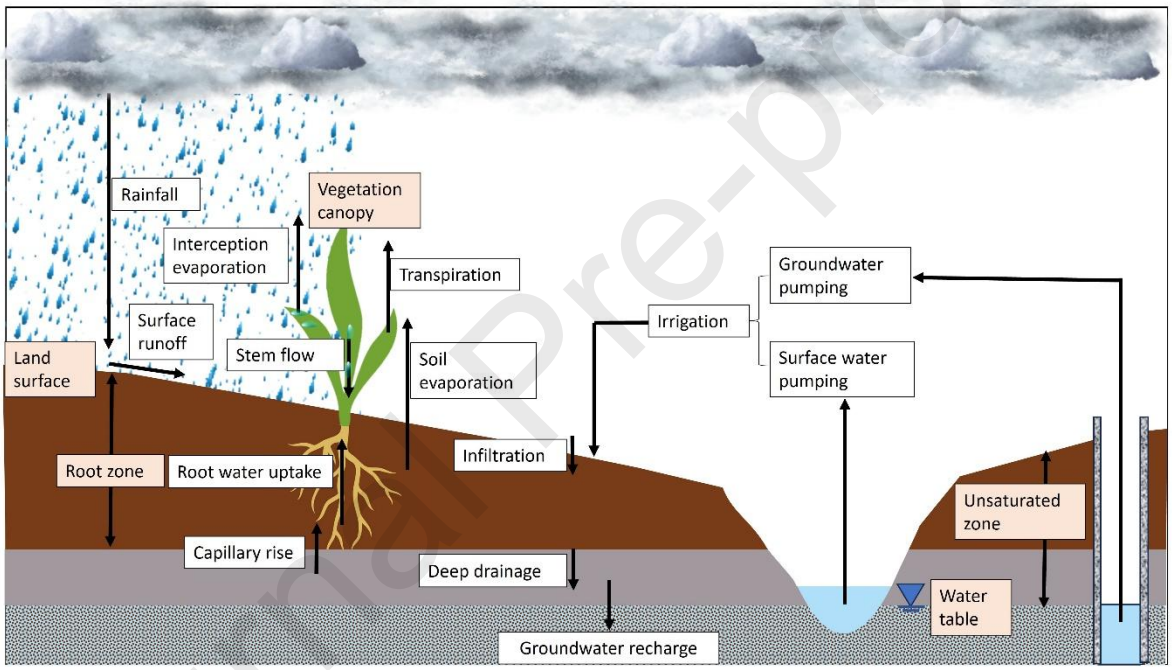
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Abstract

Soil-crop hydrology models are frequently used to estimate regional-scale groundwater recharge, crop water use, runoff and freshwater extraction rates to support irrigation demands. Because interactions between soils, crops and irrigation practices influence recharge, crop water requirements and pumping, it is important to apply models that can simulate these processes simultaneously. This review examines 16 soil-crop hydrology models and 13 case study applications in which recharge, irrigation demand and extraction rates (i.e., from surface water and/or groundwater) were estimated to support regional water accounting and water-resource investigations. Model functions for runoff, soil water movement, root water uptake, crop growth and irrigation were compared to identify differences between models and their limitations. This review showed that 13 of the 16 models incorporate crop growth simulation linked to soil water dynamics, while 7 also account for salinity stress. Twelve models allow irrigation to be triggered based on soil water content, with 4 also allowing irrigation to respond to crop stress factors. The reliability of soil-crop hydrology models for estimating pumping rates remains uncertain because pumping was validated against metered data in only 2 of the 13 case studies reviewed. In those studies, metered extraction data provided important constraints on model parameterisation. Although runoff was simulated, it was reconciled with field observations in only 1 of the 13 case studies. Soil-crop hydrology models remain important tools for estimating recharge and irrigation demand, however, the representation of conjunctive use of groundwater and surface water, and the resulting pumping estimates, are commonly imposed externally and remain weakly validated against field observations.

Highlights

- 16 models evaluated for crop, soil hydrology and irrigation simulation.
- 13 models simulate water stress; only 7 also represent salinity stress.
- Only 4 models trigger irrigation using crop-stress factors.
- Modelled pumping was compared with metered data in only 2 studies.
- Runoff was reconciled with field data in only 1 of 13 case studies.



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